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Fuzzy-driven broad learning system with class probability and density awareness for multi-view data

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ABSTRACT

The broad learning system (BLS) has recently emerged as a promising alternative to deep neural networks, offering comparable predictive accuracy with drastically reduced computational cost through random weight assignment and pseudo-inverse-based training. Despite its efficiency, conventional BLS suffers from a critical limitation: it treats all training samples equally, making it highly sensitive to outliers, label noise, and class imbalance, ubiquitous challenges in real-world data. To overcome these shortcomings, we propose a new framework, the class probability-based bell-shaped BLS (CPBS-BLS). In CPBS-BLS, each sample is assigned an adaptive membership value through a bell-shaped weighting function conditioned jointly on class probability and imbalance ratio. This mechanism automatically emphasizes reliable samples in dense regions while suppressing the influence of outliers and noisy points in sparse regions, leading to a more robust decision boundary. Although CPBS-BLS substantially improves robustness in single-view settings, its reliance on a single perspective restricts its ability to capture complementary structures present in complex, multi-faceted data. To address this, we further advance the framework by introducing the class probability-based bell-shaped multi-view BLS (CPBS-MvBLS). CPBS-MvBLS integrates multiple views of data while preserving the adaptive weighting scheme, enabling the model to exploit complementary information across diverse feature spaces. This joint treatment of sample reliability, class imbalance, and multi-view consistency represents a novel step forward in the BLS paradigm. We provide rigorous theoretical analysis establishing generalization bounds for both CPBS-BLS and CPBS-MvBLS, guaranteeing their reliability. Extensive experiments on benchmark datasets from UCI, KEEL, and AWA demonstrate that our models consistently outperform state-of-the-art baselines in accuracy, robustness to noise, and handling of imbalanced classes, as validated through statistical significance testing. The source code of the proposed CPBS-BLS and CPBS-MvBLS models are available at <https://github.com/mtanveer1/CPBS-BLS>.

1. Introduction

Broad learning systems (BLSs) (Chen & Liu, 2018) offer an efficient alternative to deep neural networks by employing a flat-broader network expansion, where randomly initialized feature nodes and enhancement nodes are incrementally incorporated to enable rapid model training. Unlike deep learning models requiring backpropagation, BLS computes output weights through pseudo-inverse method, enabling rapid training while avoiding gradient vanishing issues. The system maps input data through random feature transformation, generates enhancement nodes via nonlinear activation, and concatenates these layers for output prediction. BLS has demonstrated strong performance in time-series forecasting (Li & Law, 2024), image classification (Yi et al., 2024), medical diagnosis (Han et al., 2022) and industrial fault detection (Pu & Li, 2021). Additionally, compared to deep learning models (Feng et al.,

2022), BLS can achieve competitive accuracy with significantly reduced training time, even when the available training data is limited (Yu & Zhao, 2020).

The standard BLS architecture (Gong et al., 2021) has been extended through numerous variants to tackle a wide range of real-world challenges, each addressing specific limitations of the BLS framework. For instance, to enhance model stability and sparsity, the huber-type robust BLS (HR-BLS) (Zhang et al., 2025b) integrates the huber loss function. Meanwhile, in scenarios that require refined label learning, the groupwise label enhancement BLS (Jin et al., 2025) jointly optimizes intra-class consistency and inter-class discriminability, offering a more structured approach to classification tasks. The universal approximation capability of the BLS is proven by Chen et al. (2019), with further analysis that its structural variants maintain approximation power while offering improved representational flexibility. Beyond traditional

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learning paradigms, BLS has also been adapted for graph-based data, such as the broad graph attention network with a multikernel mechanism (BGAT-MK) (Wang et al., 2025), which is proposed to mitigate the feature oversmoothing problem prevalent in conventional graph neural networks (GNNs). In healthcare applications, Ao et al. (2024) introduced a heart failure prediction framework that captures complex patterns in medical datasets to improve predictive accuracy. To further enhance robustness, Sajid et al. (2024) introduced the intuitionistic fuzzy BLS (IF-BLS), which incorporates an intuitionistic fuzzy scheme to handle data uncertainty more effectively. BLS via adaptive maximum weighted correntropy (AMWC-BLS) (Wang et al., 2025) is designed to enhance robustness against noise and outliers while maintaining efficient learning performance. Together, these advancements demonstrate the adaptability of the BLS framework across different domains, showcasing its ability to address various real-world machine learning problems. While BLS demonstrates notable computational efficiency and effective learning, the framework possesses several intrinsic limitations that constrain its performance. Due to its flat architecture, BLS tends to overlook subtle and intricate relationships within complex datasets. Furthermore, BLS struggles with comprehensive data understanding due to its single-view processing framework, potentially leading to performance degradation and limited ability to capture underlying data relationships. To address these limitations, Huang et al. (2024) developed a multi-view BLS framework that significantly leverages the inherent relationships embedded within the datasets. Multi-view BLS improves data comprehension by simultaneously processing diverse feature representations, enabling the model to capture complementary information from multiple data perspectives and mitigate information loss inherent in single-view approaches. This advancement allows for more robust feature detection while maintaining BLS's computational advantages.

Multi-view learning has evolved into several specialized paradigms, each offering unique solutions for integrating information from multiple data perspectives. At the classifier level, support vector machines (SVMs) were extended through multi-view twin parametric margin formulations (MvTPMSVM) to avoid computationally expensive matrix inversions (Quadir & Tanveer, 2024). Quadir et al. (2025a) proposed multiview SVM with wave loss function (Wave-MvSVM), a multiview learning framework that exploits the wave loss function to integrate consensus and complementarity for enhanced robustness and synergy. Building on this, recent works extended multi-view learning to randomized neural architectures. For instance, graph random vector functional link based on multi-view learning (GRVFL-MV) (Tanveer et al., 2025) integrates graph embeddings into multi-view RVFL to preserve geometric relationships while retaining the speed of traditional RVFL networks. Similarly, multi-view extensions of RVFL have demonstrated notable performance in domain-specific tasks like DNA-binding protein prediction (Quadir et al., 2024). For clustering tasks, advanced tensor approximations in multi-view spectral clustering (MVSC) improved neighbor-preservation in high-dimensional spaces (Peng et al., 2024). Deep learning approaches like CADD-DCCNN (Li et al., 2024) introduce cross-attention with causal convolutions for multi-view EEG-based emotion recognition. In recommendation systems, MvStHgL (An et al., 2024) uses hypergraph learning to model spatial-temporal periodic patterns in point of interest (POI) data. Collectively, these models demonstrate the versatility of multi-view learning in extracting robust features and improving performance across varied domains.

Although multi-view BLS (MvBLS) models excel at integrating complementary information from diverse perspectives, their performance deteriorates significantly in the presence of noisy samples, outliers, and class imbalance. This vulnerability arises from the flat architecture of BLS, which lacks deep hierarchical feature extraction, and from the uniform treatment of all samples during training, regardless of their reliability. In practice, this means that noisy or anomalous samples exert the same influence on the model as clean, reliable ones, leading

to skewed feature representations and degraded generalization performance. Specifically:

- When noisy or anomalous features propagate through the feature and enhancement nodes of BLS, they contaminate the higher-level representations, causing distortion in the learned decision boundaries.
- Because BLS assigns equal weights to every sample, it fails to discriminate between informative samples and unreliable ones, amplifying the impact of noise, outliers, and majority-class dominance in imbalanced datasets.

To address these weaknesses, we incorporate the generalized bell-shaped membership function (Pan et al., 2021) into the BLS framework, enhancing it with class probability and imbalance ratio to adaptively evaluate the reliability of each sample. In this design, samples located near class centroids and within dense homogeneous regions are assigned higher membership values, while outliers and uncertain points in sparse or inconsistent areas are smoothly down-weighted. The class probability term reduces the influence of mislabeled or inconsistent samples, and the imbalance ratio ensures fair weighting across minority and majority classes. Through this integration, the model effectively suppresses the impact of noisy and imbalanced samples while preserving the discriminative power of clean and informative data. Fuzzy membership weighting has been extended into various specialized modeling frameworks. For instance, Dorzhigulov and James (2019) introduced a generalized bell-shaped membership-function generator implemented via memristive neural network, demonstrating that hardware-driven membership function realization can substantially improve efficiency and analog precision in neuro-fuzzy systems. Obianyo et al. (2023) applied both artificial neural networks and a neuro-fuzzy (ANFIS) model to construction scheduling, showing that the neuro-fuzzy system with Gaussian membership functions outperformed the ANN in predicting earned value metrics. Deep belief network with fuzzy parameters (Shukla & Muhuri, 2025) embeds adaptive membership functions directly into network weights, demonstrating how fuzzy parameterization can enhance learning stability and predictive performance. Similarly, Lima et al. (2025) systematically compared different membership function shapes (triangular, gaussian, bell-shaped) and defuzzification methods in a fuzzy inference system for air-quality assessment, emphasizing the sensitivity of system outputs to membership function selection. In addition, hardware-oriented research proposed an optimized field programmable gate array (FPGA)-based implementation of the bell-shaped membership function (Habeeba & Abbasb, 2025), significantly reducing hardware resource usage while maintaining functional accuracy. Collectively, these studies highlight the emerging importance of membership function design for reliable algorithms, practical applications, and hardware implementations in modern neuro-fuzzy and hybrid machine learning frameworks.

However, most existing fuzzy or multi-view BLS frameworks employ fixed or distance-based membership functions that do not adequately capture local data distribution or class uncertainty. Such approaches struggle to suppress noisy and imbalanced samples effectively, which can degrade robustness and generalization. This gap motivates the design of a more adaptive and distribution-aware weighting strategy. To bridge this gap, we propose two novel frameworks:

- CPBS-BLS, which enhances single-view BLS by integrating class probability and imbalance-aware bell-shaped weighting, thereby improving robustness to noise and imbalance while maintaining computational efficiency.
- CPBS-MvBLS, which extends this mechanism to multi-view learning. By leveraging complementary information across multiple feature spaces, CPBS-MvBLS not only captures richer underlying relationships but also ensures consistent suppression of noisy and imbalanced samples across all views.

This dual contribution introduces a reliability-driven, probability-aware, and imbalance-sensitive extension of both single-view and multi-

view BLS, addressing long-standing weaknesses of the original framework while advancing its capacity to handle complex, noisy, and imbalanced real-world datasets.

The key highlights of this paper are as follows:

1. We present a novel CPBS-BLS framework, which integrates the generalized bell-shaped membership function with class probability and imbalance ratio to adaptively weight training samples according to their reliability.
2. By evaluating each sample's proximity to class centers and local neighborhood density, CPBS-BLS emphasizes informative samples in dense regions while systematically suppressing the influence of outliers and uncertain points in sparse or distant regions.
3. We further extend this idea to multi-view learning and propose the CPBS-MvBLS model, which leverages multiple feature representations from diverse perspectives to achieve a deeper and more comprehensive understanding of data.
4. The CPBS-MvBLS framework jointly addresses outliers, label noise, and class imbalance while exploiting complementary cross-view information, thereby enhancing robustness and significantly improving predictive accuracy on complex, noisy datasets.
5. Extensive evaluations are conducted: CPBS-BLS is tested on UCI and KEEL datasets, while CPBS-MvBLS is validated on UCI, KEEL, AwA, and MNIST datasets. In addition to performance benchmarking, we provide statistical tests, theoretical generalization analysis, and hyperparameter sensitivity studies, confirming the reliability and scalability of the proposed models.

The remainder of this paper is organized as follows: [Section 2](#) reviews previous works and notations used in this paper. [Section 3](#) provides the detailed mathematical formulation of the proposed CPBS-BLS and CPBS-MvBLS frameworks. [Section 4](#) discuss the theoretical analyses on generalization bounds of CPBS-BLS and CPBS-MvBLS. Comprehensive experimental results with statistical analyses are presented in [Section 5](#). [Section 6](#) explores the impact of hyperparameter settings on model performance. [Section 7](#) presents an ablation study on weighting schemes. Finally, [Section 8](#) concludes the paper and discusses the future research directions.

2. Preliminary work

This section introduces the key mathematical notation used throughout the paper. We first provide a concise overview of the BLS ([Chen & Liu, 2017](#)) framework and its associated optimization problem. Next, we briefly describe the generalized bell-shaped membership function and its role in class probability.

2.1. Notations

In the single-view setting, the training set is represented as $\{(a_1, b_1), (a_2, b_2), \dots, (a_M, b_M)\}$, where a_r denotes the r^{th} training sample and b_r is the corresponding label vector. The complete input data can be arranged into a matrix $A \in \mathbb{R}^{M \times L}$, with each of the M rows containing L features. In the case of the proposed CPBS-MvBLS model, the training set is instead expressed as $\{(a_1^1, a_1^2, b_1), (a_2^1, a_2^2, b_2), \dots, (a_M^1, a_M^2, b_M)\}$, where a_r^1 and a_r^2 denote the r^{th} sample from view-1 and view-2, respectively. Correspondingly, the input data from each view are stored in matrices $A_1 \in \mathbb{R}^{M \times l_1}$ and $A_2 \in \mathbb{R}^{M \times l_2}$, with l_1 and l_2 representing the number of features in each view. For both single-view and multi-view formulations, the labels are collected in a matrix $B \in \mathbb{R}^{M \times 1}$, where M denotes the total number of training samples.

2.2. Broad learning system (BLS)

The BLS ([Chen & Liu, 2017](#)) is a randomized neural network architecture that expands its structure flatly and more broadly by constructing

feature layers and enhancement layers side by side, rather than stacking layers hierarchically as in deep networks. The BLS architecture consists of three primary components: feature nodes generation, enhancement nodes generation, and output layer integration. Given an input dataset A , and the transformation ψ_r , the feature nodes are generated through a series of feature transformations. The r^{th} feature group can be generated using the following mathematical equation:

$$Z_r = \psi_r(A\Omega_r^Z + \beta_r^Z) \in \mathbb{R}^{M \times n_Z}, \quad r = 1, 2, \dots, m_Z, \quad (1)$$

where Ω_r^Z and β_r^Z are randomly generated weight and bias matrices for the feature layer, respectively. n_Z is the number of nodes in each feature group. Concatenate all the m_Z feature groups to form the following feature layer Z :

$$Z = [Z_1, Z_2, \dots, Z_{m_Z}] \in \mathbb{R}^{M \times m_Z n_Z}. \quad (2)$$

Construct the j^{th} group of enhancement nodes using the following equation:

$$H_j = \phi_j(Z\Omega_j^H + \beta_j^H) \in \mathbb{R}^{M \times n_H}, \quad j = 1, 2, \dots, m_H, \quad (3)$$

where Ω_j^H and β_j^H are randomly generated weight and bias matrices for the j^{th} enhancement layer, respectively. ϕ_j is the nonlinear activation function and n_H is the number of nodes in each enhancement layer. The enhancement layer of m_H groups can be computed as:

$$H = [H_1, H_2, \dots, H_{m_H}] \in \mathbb{R}^{M \times m_H n_H}. \quad (4)$$

The final layer feature-enhancement representation is $G = [Z, H] \in \mathbb{R}^{M \times (m_Z n_Z + m_H n_H)}$, which is obtained by concatenating the feature layer Z and enhancement layer H . Therefore, the final output for BLS is formulated as:

$$B = G\Omega, \quad (5)$$

where Ω is the output weight matrix, obtained via pseudo-inverse to ensure fast training: $\Omega = G^\dagger B$, here $G^\dagger$ denotes the Moore-Penrose pseudo inverse of the concatenated matrix G .

2.3. Class probability based bell shaped (CPBS) membership function

The CPBS membership function ([Kumari et al., 2024](#)) is a fuzzy set membership function in fuzzy logic and fuzzy systems, offering enhanced adaptability in fuzzy modeling. The CPBS function is formulated by combining the class probability and the imbalance ratio with the bell-shaped membership function. This hybrid approach leverages probabilistic class information to enhance the interpretability and adaptability of the membership function, making it particularly suitable for classification and fuzzy inference systems. The CPBS membership function plays a key role in both proposed models by assigning weights to the individual samples based on their positions in higher-dimensional space. The CPBS membership function prioritizes the samples residing in dense regions to strengthen their impact during training, whereas it de-emphasizes samples distant from class centroids or located in sparse areas to reduce the influence of noise. These membership values help distinguish outliers and border samples within the majority class. Mathematically, the bell function is defined as follows:

$$\mathcal{M}_b(a_i) = \begin{cases} \frac{1}{1 + \left(\frac{\|\mathcal{K}(a_i) - Cen_+\|}{\omega} \right)^{2\tau}}, & \text{if } b_i = +1, \\ \frac{1}{1 + \left(\frac{\|\mathcal{K}(a_i) - Cen_-\|}{\omega} \right)^{2\tau}}, & \text{if } b_i = -1, \end{cases} \quad (6)$$

where $\mathcal{K}(\cdot)$ is a kernel function from the feature space to a higher-dimensional space. $Cen_+ = \frac{1}{M_+} \sum_{b_i=+1} (a_i)$ and $Cen_- = \frac{1}{M_-} \sum_{b_i=-1} (a_i)$ represent centers of positive and negative classes, where M_+ and M_- denote the total number of samples in the positive class and negative class, respectively. ω denotes the width parameter and τ corresponds shape parameter influencing the steepness of the membership curve.

The class probability for a given data point reflects the density of its own class in a local neighborhood. Data points with a high class

probability exhibit strong alignment with their respective class, indicating greater belongingness. In contrast, points with low class probability values are flagged as potential noise or outliers, reflecting a weaker class affiliation. The class probability can be computed as follows:

$$\mathcal{M}_p(a_i) = \frac{|\{a_r : \|\mathcal{K}(a_r) - \mathcal{K}(a_i)\| \leq \epsilon, b_r = b_i\}|}{|\{a_r : \|\mathcal{K}(a_r) - \mathcal{K}(a_i)\| \leq \epsilon\}|}, \quad (7)$$

where ϵ is a fixed positive real number.

Finally, the mathematical formula for the CPBS membership function can be expressed as:

$$\mathcal{M}(a_i) = \begin{cases} \mathcal{M}_b(a_i) \times \mathcal{M}_p(a_i) \times \frac{1}{m}, & \text{if } a_i \in \text{major class,} \\ 1, & \text{if } a_i \in \text{minor class,} \end{cases} \quad (8)$$

where m denotes the imbalance ratio, computed as $\frac{M_+}{M_-}$. Finally, the score matrix S is defined as:

$$S = \text{diag}(\mathcal{M}(a_1), \mathcal{M}(a_2), \dots, \mathcal{M}(a_M)). \quad (9)$$

2.4. Multi-view learning

Single-view learning models often struggle to generate reliable representations in the presence of noise, since a single feature representation may not capture the full complexity of the underlying data distribution. This limitation makes single view models more susceptible to losing discriminative patterns and the model's performance when faced with noise or outliers. These challenges underscore the need for more comprehensive approaches that can mitigate the vulnerability of single-view models to data imperfections. Multi-view feature extraction offers a powerful alternative, where each view corresponds to a distinct feature representation of the input data, which may originate from feature extraction methods such as principal component analysis. Training models on view-specific features enables the identification of discriminative patterns while reducing sensitivity to noise or distortions within individual views. Consequently, multi-view learning achieves more robust and accurate pattern recognition compared to traditional single-view approaches.

Recent advances in multi-view learning have led to significant developments across multi-view clustering, classification, and dimensionality reduction. In the context of multi-view clustering (IMC), low-rank tensor constrained multiview subspace clustering (LT-MSc) (Zhang et al., 2015) is proposed to introduce a low-rank tensor constraint to explore the complementary information from multiple views. Chao et al. (2024) proposed an incomplete contrastive multi-view clustering model with high-confidence guiding (ICMVC) to deal the multi-view data with missing values. The Fast IMC with view-independent anchors (FIMVC-VIA) model (Liu et al., 2024) is proposed to achieve faster and view-independent clustering. Structured multi-view k-means clustering (Zhang et al., 2025c) proposes a multi-view clustering method that can uncover clusters in arbitrary sizes and densities. Beyond clustering, multi-view classifiers provide robust approaches for classification problems. For example, Ahamed and Cheng (2025) proposed a multi-view time series classification model to capture patterns with properties like shift equivariance. Incomplete multi-view multi-label classification (iMvMLC) (Xie et al., 2025) is proposed to effectively handle incompleteness in views and labels and disentangle view-consistent and view-specific representations. GrainBrain (Fan et al., 2025), a robotic vision-based system, is proposed to leverage multi-view imaging and unsupervised anomaly detection with contrastive learning for automated grain quality inspection. A bi-level unsupervised multi-view graph representation learning framework (UMGRL) (Mo et al., 2025) with dual weight-net is proposed to adaptively capture node, graph, and edge importance for improved representation learning. Additionally, in the context of multi-view dimensionality reduction, Qian et al. (2014) proposed a semi-supervised dimensionality reduction framework tailored

for multi-label and multi-view learning settings. Zhang et al. (2017) proposed a multi-view dimensionality co-reduction method to preserve locality within views and consistency across views while generating flexible low-dimensional projections. An unsupervised multi-view dimensionality reduction learning framework with hierarchical sparse feature selection (MvDRHSFS) (Guo et al., 2023) is proposed to enhance representation learning by hierarchically selecting informative views and features. TCH (Zhang et al., 2025a) presents a multi-view dimensionality reduction method based on a triple contrastive head (sample level, feature level, and recovery level) that learns compact, discriminative embeddings while reducing redundancy and preserving view-specific information. These contributions underscore the adaptability of multi-view learning for clustering, classification, and representation learning.

3. Proposed work

This work proposes two novel models, CPBS-BLS and CPBS-MvBLS, to address the inherent challenges of real-world datasets, which are often characterized by noise, outliers, and severe class imbalance. Such imperfections can significantly distort the learning process, as traditional models treat all samples equally, allowing unreliable data points to disproportionately influence the decision boundaries. To mitigate this issue, the CPBS-BLS model incorporates a class probability-based bell-shaped (CPBS) membership function that adaptively weights each sample. By assigning higher importance to dense, representative data points while down-weighting anomalous or uncertain ones, CPBS-BLS achieves improved robustness and enhanced generalization even under impure data conditions (Quadri et al., 2025b).

While CPBS-BLS effectively strengthens learning in single-view settings, its reliance on a single feature representation restricts its ability to capture the full complexity of real-world data. Many practical datasets are inherently multi-faceted, where different feature spaces may contain complementary information. To overcome this limitation, we extend CPBS-BLS into a multi-view framework, resulting in the CPBS-MvBLS model. By integrating multiple feature perspectives, CPBS-MvBLS provides a more holistic characterization of data structure and captures richer inter-feature dependencies. Furthermore, the CPBS membership function is embedded within the multi-view framework, ensuring that noisy or misleading samples are consistently suppressed across views, while informative patterns are adaptively emphasized.

Together, the CPBS-BLS and CPBS-MvBLS models offer a unified solution to the dual challenges of noise resilience and multi-view learning. CPBS-BLS ensures stable and robust learning in single-view scenarios, whereas CPBS-MvBLS extends these advantages to multi-view contexts, enabling superior representation learning in complex, real-world applications. The following subsections provide detailed formulations of both models.

3.1. Class probability based bell shaped - broad learning system (CPBS-BLS)

The mathematical formulation and architectural design of the proposed CPBS-BLS model (see Fig. 1) are presented in this subsection. The central idea of CPBS-BLS lies in adaptively weighting training samples according to their reliability using the CPBS membership function. By emphasizing trustworthy samples while suppressing the influence of noisy or outlier data, the model achieves improved stability and robustness.

To estimate the output weights, we formulate an optimization problem that integrates both sample-specific membership values and an L_2 regularization term. The membership values ensure that each sample contributes proportionally to its estimated reliability, while the regularization term mitigates overfitting by controlling the model's structural complexity. Together, these components enable the model to minimize reconstruction error, reduce sensitivity to data impurities, and enhance

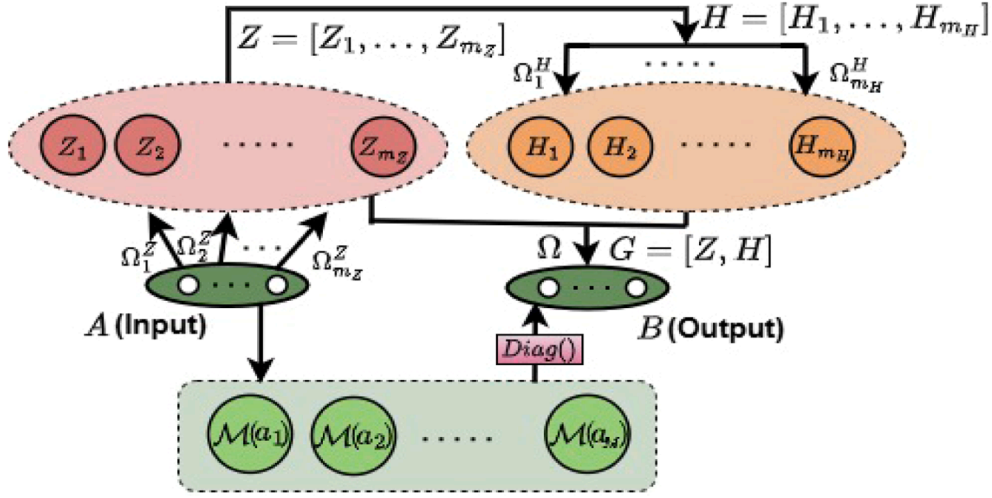


Fig. 1. Architecture of the CPBS-BLS model.

the predictive performance of the BLS framework. The resulting optimization problem of the CPBS-BLS model is expressed as:

$$\arg \min_{\Omega} \frac{1}{2} \|\Omega\|_2^2 + \frac{c}{2} \|S\xi\|_2^2 \quad (10)$$

s.t. $G\Omega - B = \xi,$

where S represents the score matrix computed from Section 2.3. ξ is the error, c is the regularization parameter, and G denotes the concatenation of the feature and enhancement layer, defined as: $G = [Z, H] \in \mathbb{R}^{M \times (m_Z n_Z + m_H n_H)}$. The Lagrangian $\mathcal{L}_B$ of the optimization problem (10) is given as:

$$\mathcal{L}_B = \frac{1}{2} \|\Omega\|_2^2 + \frac{c}{2} \|S\xi\|_2^2 - \alpha'(G\Omega - B - \xi). \quad (11)$$

By partially differentiating the Lagrangian w.r.t. Ω , ζ , α and putting them equal to zero, we obtain the following equations:

$$\Omega - G^t \alpha = 0, \quad (12)$$

$$c S^T S \xi + \alpha = 0, \quad (13)$$

$$G\Omega - B - \xi = 0. \quad (14)$$

We find the output matrix Ω , by solving Eqs. (12), (13), and (14) as below:

$$\Omega = \left(\frac{I}{c} + G^t S^t S G \right)^{-1} G^t S^t S B, \quad (15)$$

where I represents the identity matrix of suitable order. Again by solving Eqs. (12), (13), and (14) in different way, we get

$$\Omega = G^t \left(\frac{I}{c} + S^t S G G^t \right)^{-1} S^t S B. \quad (16)$$

Thereby, we derive two distinct expressions for computing Ω , given by Eqs. (15) and (16). Both formulations involve matrix inversion, but their applicability depends on the relationship between the feature dimensions ($m_Z n_Z + m_H n_H$) and the number of samples (M). Specifically, if the feature dimension ($m_Z n_Z + m_H n_H$) is less than or equal to the sample size (M), we compute Ω using Eq. (15). Otherwise, Ω will be determined using the Eq. (16). This approach provides a computational advantage by allowing matrix inversion to be performed in either the feature space or the sample space, depending on the problem's dimensionality. Thus, the optimal solution to (10) can be given by:

$$\Omega = \begin{cases} \left(G^t S^t S G + \frac{I}{c} \right)^{-1} G^t S^t S B, & \text{if } (m_Z n_Z + m_H n_H) \leq M, \\ G^t \left(S^t S G G^t + \frac{I}{c} \right)^{-1} S^t S B, & \text{if } M < (m_Z n_Z + m_H n_H). \end{cases} \quad (17)$$

The algorithm of the proposed CPBS-BLS model is briefly described in Algorithm 1.

Algorithm 1 Proposed CPBS-BLS model.

- 1: **Input:** The training dataset A , the output matrix B , the hyperparameter c , membership function parameter τ and ω .
- 2: **Input Sample:** Take a_i as the input sample, where $i = 1, 2, \dots, M$.
- 3: **Find:** The feature groups $Z_r \in \mathbb{R}^{M \times n_Z}$ using the Eq. (1).
- 4: **Find:** The feature layer $Z \in \mathbb{R}^{M \times m_Z n_Z}$ using the Eq. (2).
- 5: **Find:** The enhancement groups $H_j \in \mathbb{R}^{M \times n_H}$ using the Eq. (3).
- 6: **Find:** The enhancement layer $H \in \mathbb{R}^{M \times m_H n_H}$ using the Eq. (4).
- 7: **Compute:** The concatenated feature-enhancement matrix $G = [Z, H] \in \mathbb{R}^{M \times (m_Z n_Z + m_H n_H)}$.
- 8: **Compute:** The bell membership $\mathcal{M}_b(a_i)$, using Eq. (6).
- 9: **Compute:** The class probability $\mathcal{M}_p(a_i)$, using the Eq. (7).
- 10: **Compute:** The CPBS membership value $\mathcal{M}(a_i)$, using Eq. (8).
- 11: **Compute:** The score matrix S using the Eq. (9).
- 12: **Output:** Use Eq. (17) to obtain the output weight matrix Ω .

3.2. Class probability based bell shaped - multi-view broad learning system (CPBS-MvBLS)

The CPBS-BLS model has proven highly effective in handling noisy data by adaptively weighting each sample according to its reliability. However, its single-view design inherently limits its ability to capture complementary patterns that may exist in other feature representations. This limitation becomes especially pronounced in complex, multi-modal datasets, where diverse feature spaces often contain unique and informative patterns. To address this shortcoming, we propose the CPBS-MvBLS model, a multi-view extension of CPBS-BLS. By integrating multiple feature perspectives, CPBS-MvBLS provides a more comprehensive understanding of the underlying data structure. The model further leverages the CPBS membership function to prioritize the most informative features across views while suppressing the impact of noise and impurities. This adaptive combination of multi-view learning and CPBS membership enables the model to detect robust latent patterns even in the presence of outliers or class imbalances.

The CPBS-MvBLS optimization problem incorporates L_2 regularization for both views to reduce overfitting. To minimize prediction error, the formulation includes membership-weighted error terms for each view, ensuring that more reliable samples have greater influence during training. Additionally, a membership-weighted coupling term is introduced to jointly minimize errors across views while maintaining the relative contribution of each sample. By combining these components, CPBS-MvBLS achieves enhanced data representation, improved robustness to outliers, and superior predictive performance compared to con-

ventional approaches on imbalanced or noisy datasets. The mathematical formulation of the CPBS-MvBLS optimization problem is presented below:

$$\arg \min_{\Omega_1, \Omega_2} \frac{c_1}{2} \|S_1 \xi_1\|_2^2 + \frac{c_2}{2} \|S_2 \xi_2\|_2^2 + \frac{c_3}{2} \|\Omega_1\|_2^2 + \frac{1}{2} \|\Omega_2\|_2^2 + \rho (S_1 \xi_1)' (S_2 \xi_2) \quad (18)$$

s.t. $G_1 \Omega_1 - B = \xi_1$ and $G_2 \Omega_2 - B = \xi_2$,

where $G_1 = [Z_1, H_1]$ and $G_2 = [Z_2, H_2]$ are the concatenated feature-enhancement matrices for view-1 and view-2, respectively. Z_1 (Z_2) is the feature layer and H_1 (H_2) is the enhancement layer for view-1 (view-2). S_1 (S_2) denotes the score matrix corresponding to view-1 (view-2). The weight matrices Ω_1 and Ω_2 connect the concatenated layers G_1 and G_2 to the output layer B . To enhance the model's performance, a coupling term $(S_1 \xi_1)' (S_2 \xi_2)$ is incorporated into (18). This term facilitates the preservation of an optimal membership distribution for input samples while simultaneously minimizing errors across both views. Additionally, the optimization problem incorporates regularization hyperparameters c_1 , c_2 , and c_3 , along with a coupling hyperparameter ρ . The Lagrangian of the optimization problem (18) can be computed as:

$$\mathcal{L}_{MB} = \frac{c_1}{2} \|S_1 \xi_1\|_2^2 + \frac{c_2}{2} \|S_2 \xi_2\|_2^2 + \frac{c_3}{2} \|\Omega_1\|_2^2 + \frac{1}{2} \|\Omega_2\|_2^2 + \rho (S_1 \xi_1)' (S_2 \xi_2) - \alpha_1' (G_1 \Omega_1 - B - \xi_1) - \alpha_2' (G_2 \Omega_2 - B - \xi_2). \quad (19)$$

By partially differentiating the Lagrangian $\mathcal{L}_{MB}$ and putting them equal to zero, we obtain the following equations:

$$\frac{\partial \mathcal{L}_{MB}}{\partial \xi_1} = c_1 S_1' S_1 \xi_1 + \rho S_1' S_2 \xi_2 + \alpha_1 = 0, \quad (20)$$

$$\frac{\partial \mathcal{L}_{MB}}{\partial \xi_2} = c_2 S_2' S_2 \xi_2 + \rho S_2' S_1 \xi_1 + \alpha_2 = 0, \quad (21)$$

$$\frac{\partial \mathcal{L}_{MB}}{\partial \Omega_1} = c_3 \Omega_1 - G_1' \alpha_1 = 0, \quad (22)$$

$$\frac{\partial \mathcal{L}_{MB}}{\partial \Omega_2} = \Omega_2 - G_2' \alpha_2 = 0, \quad (23)$$

$$\frac{\partial \mathcal{L}_{MB}}{\partial \alpha_1} = G_1 \Omega_1 - B - \xi_1 = 0, \quad (24)$$

$$\frac{\partial \mathcal{L}_{MB}}{\partial \alpha_2} = G_2 \Omega_2 - B - \xi_2 = 0. \quad (25)$$

By solving for α_1 and α_2 from (20) and (21), respectively, and substituting them into (22) and (23), we derive the analytical expressions:

$$c_3 \Omega_1 + G_1' (c_1 S_1' S_1 \xi_1 + \rho S_1' S_2 \xi_2) = 0, \quad (26)$$

$$\Omega_2 + G_2' (c_2 S_2' S_2 \xi_2 + \rho S_2' S_1 \xi_1) = 0. \quad (27)$$

The variables ξ_1 and ξ_2 are determined from (24) and (25), respectively, and subsequently substituted into (26) and (27) to obtain:

$$c_3 \Omega_1 + G_1' (c_1 S_1' S_1 (G_1 \Omega_1 - B) + \rho S_1' S_2 (G_2 \Omega_2 - B)) = 0, \quad (28)$$

$$\Omega_2 + G_2' (c_2 S_2' S_2 (G_2 \Omega_2 - B) + \rho S_2' S_1 (G_1 \Omega_1 - B)) = 0. \quad (29)$$

The Eqs. (28) and (29) can be reformulated as follows:

$$\Omega_1 (c_3 I_1 + c_1 G_1' S_1' S_1 G_1) + \Omega_2 (\rho G_1' S_1' S_2 G_2) = G_1' (c_1 S_1' S_1 + \rho S_1' S_2) B, \quad (30)$$

$$\Omega_1 (\rho G_2' S_2' S_1 G_1) + \Omega_2 (I_2 + c_2 G_2' S_2' S_2 G_2) = G_2' (c_2 S_2' S_2 + \rho S_2' S_1) B, \quad (31)$$

where I_1 and I_2 are the identity matrices of appropriate dimension. Eqs. (30) and (31) can be written in the matrix form as follows:

$$\begin{bmatrix} c_3 I_1 + c_1 G_1' S_1' S_1 G_1 & \rho G_1' S_1' S_2 G_2 \\ \rho G_2' S_2' S_1 G_1 & I_2 + c_2 G_2' S_2' S_2 G_2 \end{bmatrix} \begin{bmatrix} \Omega_1 \\ \Omega_2 \end{bmatrix} = \begin{bmatrix} G_1' (c_1 S_1' S_1 + \rho S_1' S_2) \\ G_2' (c_2 S_2' S_2 + \rho S_2' S_1) \end{bmatrix} B. \quad (32)$$

Finally, we get the expression for output weight matrices Ω_1 and Ω_2 by taking the inverse:

$$\begin{bmatrix} \Omega_1 \\ \Omega_2 \end{bmatrix} = \begin{bmatrix} c_3 I_1 + c_1 G_1' S_1' S_1 G_1 & \rho G_1' S_1' S_2 G_2 \\ \rho G_2' S_2' S_1 G_1 & I_2 + c_2 G_2' S_2' S_2 G_2 \end{bmatrix}^{-1} \begin{bmatrix} G_1' (c_1 S_1' S_1 + \rho S_1' S_2) \\ G_2' (c_2 S_2' S_2 + \rho S_2' S_1) \end{bmatrix} B. \quad (33)$$

The algorithm of the proposed CPBS-MvBLS model is briefly described in Algorithm 2.

Algorithm 2 Proposed CPBS-MvBLS model.

- 1: **Input:** The training dataset A_1 for view-1 and A_2 for view-2, the output matrix B , the parameter c_1 , c_2 , c_3 , ρ , and membership function parameter τ and ω .
- 2: **Input Sample:** Take a_1^i and a_2^i as the input samples for view-1 and view-2, where $i = 1, 2, \dots, M$.
- 3: **Find:** The feature groups for both views using the Eq. (1).
- 4: **Find:** The feature layers $Z_1 \in \mathbb{R}^{M \times m_Z n_Z}$ and $Z_2 \in \mathbb{R}^{M \times m_Z n_Z}$ for view-1 and view-2, using the Eq. (2).
- 5: **Find:** The enhancement groups for both view, using the Eq. (3).
- 6: **Find:** The enhancement layers $H_1 \in \mathbb{R}^{M \times m_H n_H}$ and $H_2 \in \mathbb{R}^{M \times m_H n_H}$, using the Eq. (4).
- 7: **Compute:** The concatenated feature-enhancement matrices $G_1 = [Z_1, H_1] \in \mathbb{R}^{M \times (m_Z n_Z + m_H n_H)}$ and $G_2 = [Z_2, H_2] \in \mathbb{R}^{M \times (m_Z n_Z + m_H n_H)}$ corresponding to view-1 and view-2, respectively.
- 8: **Compute:** The bell membership $\mathcal{M}_b(a_1^i)$ and $\mathcal{M}_b(a_2^i)$ across both the views, using the Eq. (6).
- 9: **Compute:** The class probability $\mathcal{M}_p(a_1^i)$ and $\mathcal{M}_p(a_2^i)$ across both the views, using the Eq. (7).
- 10: **Compute:** The CPBS membership value $\mathcal{M}(a_1^i)$ and $\mathcal{M}(a_2^i)$ for both the views, using the Eq. (8).
- 11: **Compute:** The score matrix S_1 (view-1) and S_2 (view-2), using the Eq. (9).
- 12: **Output:** Use Eq. (33) to obtain the output weight matrix Ω_1 for and Ω_2 .

3.3. Time complexity analysis

We analyze the time complexity of the proposed CPBS-BLS and CPBS-MvBLS models by decomposing the analysis into two components: (1) membership value computation and (2) BLS model complexity.

CPBS-BLS Complexity. Given an input dataset with M number of samples, the membership value calculation for each sample exhibits a time complexity of $\mathcal{O}(M)$ (Rezvani et al., 2019a). However, the dominant computational cost arises from computing the unknown output matrices. Specifically: if $M \geq (m_Z n_Z + m_H n_H)$, the computational complexity for finding the output matrix Ω is $\mathcal{O}((m_Z n_Z + m_H n_H)^3)$ and if $M < (m_Z n_Z + m_H n_H)$, the complexity scales as $\mathcal{O}(M^3)$. Thereby, the total time complexity is $\mathcal{O}(M) + \mathcal{O}((m_Z n_Z + m_H n_H)^3)$ or $\mathcal{O}(M) + \mathcal{O}(M^3)$ for both the cases, respectively. Since $\mathcal{O}(M)$ is negligible compared to the computational complexity for computing the output weight matrices in both cases, the overall time complexity of CPBS-BLS reduces to: $\mathcal{O}((m_Z n_Z + m_H n_H)^3)$ for $M \geq (m_Z n_Z + m_H n_H)$, and $\mathcal{O}(M^3)$ otherwise.

CPBS-MvBLS Complexity. Analogously, membership value computation requires $\mathcal{O}(M)$ time. The main complexity arises from the matrix inversion in the optimization solution (Eq. (33)), resulting in $\mathcal{O}((m_Z n_Z + m_H n_H)^3)$. The total time complexity is $\mathcal{O}(M) + \mathcal{O}((m_Z n_Z + m_H n_H)^3)$. Since $\mathcal{O}(M)$ is negligible, the time complexity simplifies to: $\mathcal{O}((m_Z n_Z + m_H n_H)^3)$.

Computational Complexity Comparison: The computational complexities of the proposed CPBS-BLS and CPBS-MvBLS models are $\mathcal{O}(\lceil \min((m_Z n_Z + m_H n_H), M) \rceil^3)$ and $\mathcal{O}((m_Z n_Z + m_H n_H)^3)$, respectively. Hence, both models exhibit cubic computational complexity. Importantly, this complexity is consistent with that of the baseline methods, as all competing approaches also demonstrate cubic time complexity. Consequently, the proposed frameworks achieve improved performance without incurring any additional computational burden. The comparative computational complexities of all methods are summarized in Table 1.

4. Theoretical analysis of the proposed models

Establishing the theoretical reliability of a learning algorithm is crucial for validating its prediction accuracy. In this section, we provide

Table 1
Computational complexities comparison across the proposed and baseline models.

Model	Time Complexity
ELM (Huang et al., 2006)	$\mathcal{O}(M^3)$
BLS (Chen & Liu, 2017)	$\mathcal{O}(M^3)$
IF-BLS (Sajid et al., 2024)	$\mathcal{O}(M^3)$
NF-BLS (Feng & Chen, 2018)	$\mathcal{O}(ML \sum_{i=1} K_i^2 + M \sum_{i=1} K_i \sum_{j=1}^{m_H} n_H + (\sum_{i=1} K_i + \sum_{j=1}^{m_H} n_H))$
F-BLS (Sajid et al., 2024)	$\mathcal{O}(M^3)$
IF-TSVM (Rezvani et al., 2019b)	$\mathcal{O}(2 \times (M/2)^3)$
H-ELM (Tang et al., 2015)	$\mathcal{O}(ML(m_Z n_Z + m_H n_H) + M(m_Z n_Z + m_H n_H)^2 + (m_Z n_Z + m_H n_H)^3)$
SVM2K (Farquhar et al., 2005)	$\mathcal{O}(M^3)$
MVLDM (Hu et al., 2025)	$\mathcal{O}(M^3)$
MvTSVM (Xie & Sun, 2015)	$\mathcal{O}(M^3)$
CPBS-BLS ^a	$\mathcal{O}(\min\{(m_Z n_Z + m_H n_H), M\}^3)$
CPBS-MvBLS ^a	$\mathcal{O}((m_Z n_Z + m_H n_H)^3)$

^a Represent the proposed models.

a theoretical analysis of the proposed models by deriving generalization bounds, which quantify the probability that the prediction error, measured using appropriate norms, remains within specified limits. This analysis offers formal guarantees on the models' performance and provides insight into their robustness and stability.

Definition 1. Let R_G define the *Stable rank* of G then $R_G = \frac{\|G\|_F^2}{\|G\|_2^2}$, where $\|\cdot\|_F$ is Frobenius norm of a matrix and $\|\cdot\|_2$ is L_2 norm of a matrix.

Definition 2. Let $V = (v_{ij})$ be an $n \times k$ order matrix such that $k \leq n$ and where each entry v_{ij} is an identically independent (i.i.d.) random variable following the standard normal distribution. The *Gaussian estimator* for the Frobenius norm of a matrix G is denoted by $\mathcal{E}(G)$ and defined as:

$$\mathcal{E}(G) = \frac{1}{\sqrt{k}} \|GV\|_F.$$

Lemma 1 (following Kim (2023)). For any $\eta > 1$, the following probabilistic bounds hold for the Gaussian estimator $\mathcal{E}(G)$:

$$P\left\{\mathcal{E}(G) \geq \eta \|G\|_F\right\} \leq \exp\left[-\frac{k}{2} R_G \left\{(1-\Gamma)(\eta^2-1) - \frac{(1-\Gamma)^2 R_G}{2\Gamma}\right\}\right], \quad (34)$$

$$P\left\{\mathcal{E}(G) \leq \frac{1}{\eta} \|G\|_F\right\} \leq \exp\left\{-\frac{k}{4} \frac{(\eta^2-1)^2}{\eta^4}\right\}, \quad (35)$$

where $\Gamma = \sqrt{\frac{R_G}{2(\eta^2-1) + R_G}}$. Consequently, we have the combined probability bound:

$$P\left\{\frac{1}{\eta} \|G\|_F < \mathcal{E}(G) < \eta \|G\|_F\right\} \geq 1 - \exp\left[-\frac{k}{2} R_G \left\{(1-\Gamma)(\eta^2-1) - \frac{(1-\Gamma)^2 R_G}{2\Gamma}\right\}\right] - \exp\left\{-\frac{k}{4} \frac{(\eta^2-1)^2}{\eta^4}\right\}. \quad (36)$$

Let Ω be an unknown weight matrix satisfying $B = G\Omega$. Let B^{out} denote the predicted output matrix and B the expected output matrix, implying a prediction error between them. Denote this error by $\mathbf{E}$. Without loss of generality, we assume that the elements of $\mathbf{E}$ are independent and identically distributed (i.i.d.) with zero mean and variance σ^2 , i.e., $\mathbf{E} \sim \mathcal{N}(0, \sigma^2 I)$, where I is the identity matrix. Under this assumption, the expected output can be expressed as:

$$B = G\Omega^{\text{out}} + \mathbf{E}, \quad (37)$$

where Ω^{out} is the predicted output weight matrix corresponding to $B^{\text{out}} = G\Omega^{\text{out}}$. We employ the L_2 norm to quantify the prediction error, denoted as $\|B - B^{\text{out}}\|_2$. Recall that the CPBS membership function $\mathcal{M}(a_i)$ combines a bell-shaped distance term, a local class-probability term, and the imbalance ratio, and that the resulting score matrix is

$S = \text{diag}(\mathcal{M}(a_1), \dots, \mathcal{M}(a_2))$. In CPBS-BLS, the least-squares fitting is not performed on the original design matrix G but on the weighted design matrix $G := SG$. Concretely, if G_i denotes the i^{th} row of G then the corresponding i^{th} row of G is $G_i := \mathcal{M}(a_i) \cdot G_i$. Thus, each sample's contribution to the gram matrix $G^T G$ and to the loss function is scaled by its membership value. This effect of the design matrix provides a direct connection between the proposed CPBS membership function and measurable statistical properties of the model.

4.1. Generalization error bound for the CPBS-BLS

In this subsection, we derive probabilistic bounds on the generalization error of the CPBS-BLS model. These bounds provide a theoretical guarantee on the prediction performance by quantifying the likelihood that the observed error deviates from its expected behavior. This analysis is crucial for understanding the stability and reliability of the model when applied to real-world noisy or imbalanced datasets. Although prediction error components are not strictly gaussian in arbitrary datasets, for randomized networks such as BLS, the concentration of random projections causes the output residuals to converge toward a normal distribution due to the central limit theorem (CLT) (see Jha and Singh (2021)). Therefore, the assumption $\mathbf{E} \sim \mathcal{N}(0, I)$ holds approximately in practice and is consistent with prior generalization analyses of BLS models.

Theorem 1. For any $\eta > 1$, the probabilistic upper bound for the deviation $\|B - B^{\text{out}}\|_2$ can be written as $P\left\{\|B - B^{\text{out}}\|_2 \geq \eta \cdot \frac{\sigma}{R_{G^\dagger E}} \|G\|_2 \|G^\dagger\|_F\right\} \leq \exp\left[-\frac{1}{2} R_{G^\dagger} \left\{(1-\Gamma^+)(\eta^2-1) - \frac{(1-\Gamma^+)^2 R_{G^\dagger}}{2\Gamma^+}\right\}\right]$, where $\Gamma^+ = \sqrt{\frac{R_{G^\dagger}}{2(\eta^2-1) + R_{G^\dagger}}}$, $G^\dagger$ is the pseudo inverse of G , whereas the lower bound is $P\left\{\|B - B^{\text{out}}\|_2 \leq \frac{1}{\eta} \cdot \frac{\sigma}{R_{G^\dagger E}} \|G\|_2 \|G^\dagger\|_F\right\} \leq \exp\left\{-\frac{1}{4} \frac{(\eta^2-1)^2}{\eta^4}\right\}$. Then, by merging the lower and upper bounds, the probability of the generalization bound for the predicted error $\|B - B^{\text{out}}\|_2$ is:

$$P\left\{\frac{1}{\eta} \cdot \frac{\sigma}{R_{G^\dagger E}} \|G\|_2 \|G^\dagger\|_F < \|B - B^{\text{out}}\|_2 < \eta \cdot \frac{\sigma}{R_{G^\dagger E}} \|G\|_2 \|G^\dagger\|_F\right\} \geq 1 - \exp\left\{-\frac{1}{4} \frac{(\eta^2-1)^2}{\eta^4}\right\} - \exp\left[-\frac{1}{2} R_{G^\dagger} \left\{(1-\Gamma^+)(\eta^2-1) - \frac{(1-\Gamma^+)^2 R_{G^\dagger}}{2\Gamma^+}\right\}\right]. \quad (38)$$

Proof.

By the Lemma 1 with $k = 1$ and normalizing $\mathbf{E}' = \mathbf{E}/\sigma \sim \mathcal{N}(0, I)$, we obtain the following inequality:

$$P\{\mathcal{E}(G^\dagger) \geq \eta \|G^\dagger\|_F\} = P\{\|G^\dagger \mathbf{E}'\|_F \geq \eta \|G^\dagger\|_F\} \leq \exp\left[-\frac{1}{2} R_{G^\dagger} \left\{(1-\Gamma^+)(\eta^2-1) - \frac{(1-\Gamma^+)^2 R_{G^\dagger}}{2\Gamma^+}\right\}\right], \quad (39)$$

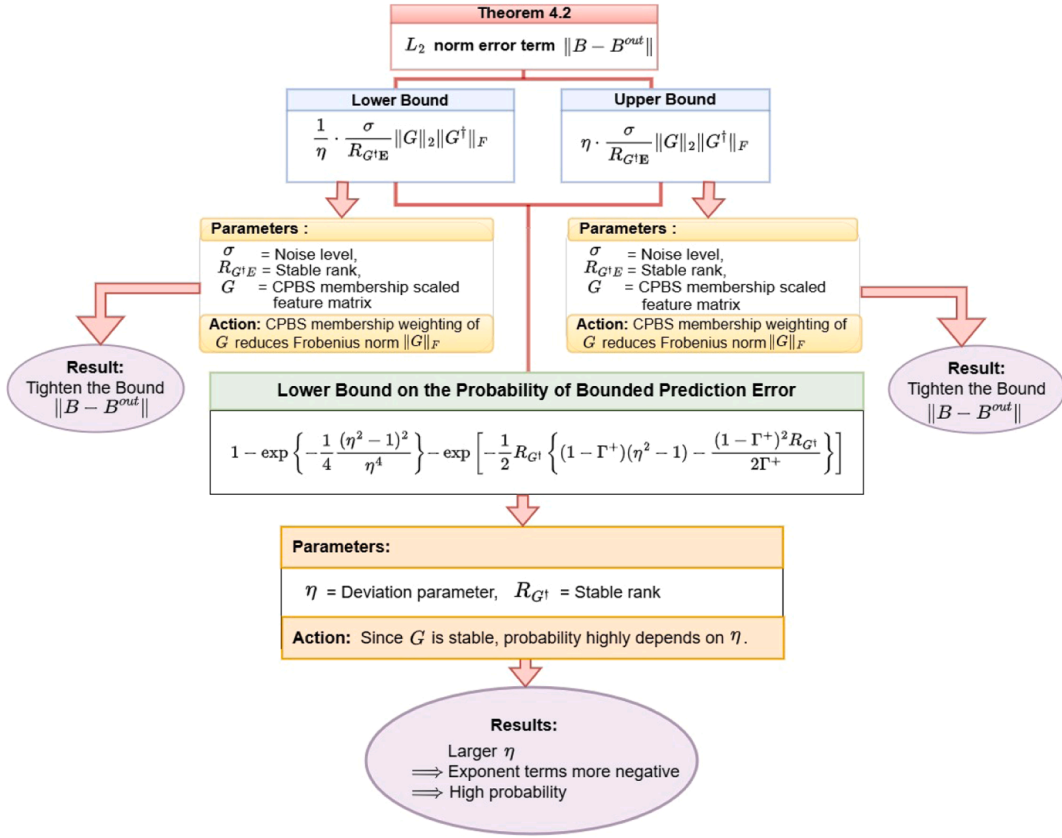


Fig. 2. Graphical representation of Theorem 1 demonstrating how spectral properties of the membership-scaled feature matrix G and deviation parameter η influence the deviation bounds and probability concentration.

Consider the L.H.S. part of the inequality (39):

$$\begin{aligned} P\{\|G^\dagger E'\|_F \geq \eta \|G^\dagger\|_F\} &= P\left\{\|G^\dagger E\|_2 \geq \eta \frac{\sigma}{R_{G^\dagger E}} \|G^\dagger\|_F\right\} \\ &= P\left\{\|\Omega - \Omega^{out}\|_2 \geq \eta \frac{\sigma}{R_{G^\dagger E}} \|G^\dagger\|_F\right\}. \end{aligned} \quad (40)$$

Eqs. (39) and (40) establish the following inequality:

$$\begin{aligned} P\left\{\|\Omega - \Omega^{out}\|_2 \geq \eta \frac{\sigma}{R_{G^\dagger E}} \|G^\dagger\|_F\right\} \\ \leq \exp\left[-\frac{1}{2} R_{G^\dagger} \left\{(1 - \Gamma^+)(\eta^2 - 1) - \frac{(1 - \Gamma^+)^2 R_{G^\dagger}}{2\Gamma^+}\right\}\right]. \end{aligned} \quad (41)$$

Similarly, for the lower bound (using Eq. (35) from Lemma 1, we obtain the following inequality:

$$P\left\{\|G^\dagger E'\|_F \leq \frac{1}{\eta} \|G^\dagger\|_F\right\} \leq \exp\left\{-\frac{1}{4} \frac{(\eta^2 - 1)^2}{\eta^4}\right\}. \quad (42)$$

Also, from the L.H.S. part of the inequality (42), we have:

$$\begin{aligned} P\{\|G^\dagger E'\|_F \leq \frac{1}{\eta} \|G^\dagger\|_F\} &= P\left\{\|G^\dagger E\|_2 \leq \frac{1}{\eta} \frac{\sigma}{R_{G^\dagger E}} \|G^\dagger\|_F\right\} \\ &= P\left\{\|\Omega - \Omega^{out}\|_2 \leq \frac{1}{\eta} \frac{\sigma}{R_{G^\dagger E}} \|G^\dagger\|_F\right\}. \end{aligned} \quad (43)$$

Hence, from Eqs. (42) and (43), we obtain:

$$P\left\{\|\Omega - \Omega^{out}\|_2 \leq \frac{1}{\eta} \frac{\sigma}{R_{G^\dagger E}} \|G^\dagger\|_F\right\} \leq \exp\left\{-\frac{1}{4} \frac{(\eta^2 - 1)^2}{\eta^4}\right\}. \quad (44)$$

Since $\|B - B^{out}\|_2 \leq \|G\|_2 \|\Omega - \Omega^{out}\|_2$ (using triangular inequality). The final result can be obtained directly by combining Eqs. (41) and (44). (Fig. 2) $\square$

4.2. Interpretation and graphical representation of Theorem 1 :

Theorem 1 provides a probabilistic generalization bound for the error between the predicted output matrix B^{out} and the expected output matrix B . The bound depends on the noise level σ , stable rank $R_{G^\dagger E}$, and the term $\|G\|_2 \|G^\dagger\|_F$. These terms directly link the conditioning and stability of the CPBS membership-scaled concatenated matrix G to the robustness of the predicted output weights in the proposed CPBS-BLS model. In particular, the deviation radius is proportional to $\|G\|_2 \|G^\dagger\|_F$ and $R_{G^\dagger E}$. The incorporation of the CPBS membership mechanism modifies the concatenated feature matrix as $G = S \cdot [Z, H]$, where S is a diagonal score matrix that reflects class probability and density awareness, Z is the feature layer, and H is the enhancement layer. This weighted formulation improves the spectral distribution of G in the feature space, thereby reducing the pseudo-inverse norm $\|G^\dagger\|_F$ and improving the effective conditioning of G . Consequently, the admissible deviation radius becomes smaller compared to traditional BLS, yielding a tighter generalization bound. Furthermore, as shown in Eq. 38, the lower bound on the probability of the bounded error increases as the deviation parameter $\eta > 1$ grows. Since the probability terms exhibit exponential decay in η , larger values of η result in more negative exponents, which guarantees that large deviations from the optimal solution become increasingly unlikely. From an intuitive perspective, Theorem 1 implies that when the final feature matrix G is well-conditioned, the predicted output weights remain close to the optimal solution with high probability. By incorporating class probability and density awareness, the CPBS-BLS model enhances the discriminative and structural properties of the feature representation. This improved spectral stability reduces the sensitivity of the pseudo-inverse solution to noise. Therefore, Theorem 1 provides a rigorous theoretical justification for the robustness and improved generalization capability of the proposed CPBS-BLS model. The exponential

concentration terms ensure statistical stability, while the membership-scaled conditioning mechanism yields a tighter deviation bound compared to traditional BLS.

4.3. Generalization error bound for the CPBS-MvBLS

The generalization bound for the proposed CPBS-MvBLS model can be obtained by extending [Theorem 1](#) to the multi-view scenario. Specifically, by replacing the single-view matrix G with the concatenated weight matrix $[\Omega_1, \Omega_2]$ as defined in [Eq. \(18\)](#), the probabilistic framework naturally accounts for contributions from multiple feature perspectives. This adaptation ensures that the theoretical guarantees on prediction error and model stability remain valid, demonstrating that CPBS-MvBLS can effectively maintain robust generalization even when integrating information across multiple views.

5. Implementation and evaluation

Implementation Details. In this section, we employ an extensive assessment of the proposed CPBS-BLS and CPBS-MvBLS models, benchmarking the performance of CPBS-BLS against BLS ([Chen & Liu, 2018](#)), extreme learning model (ELM) ([Huang et al., 2006](#)), intuitionistic fuzzy-BLS (IF-BLS) ([Sajid et al., 2024](#)), neuro fuzzy BLS (NF-BLS) ([Feng & Chen, 2018](#)), fuzzy BLS (F-BLS) ([Sajid et al., 2024](#)), intuitionistic fuzzy twin support vector machine (IF-TSVM) ([Rezvani et al., 2019b](#)), hierarchical ELM (H-ELM) ([Tang et al., 2015](#)), and deep ensemble RVFL (edRVFL) ([Shi et al., 2021](#)), whereas the proposed CPBS-MvBLS model compared against ELM1 (ELM on view-1) ([Huang et al., 2006](#)), ELM2 (ELM on view-2) ([Huang et al., 2006](#)), BLS1 (BLS on view-1) ([Chen & Liu, 2018](#)), BLS2 (BLS on view-2) ([Chen & Liu, 2018](#)), SVM2K ([Farquhar et al., 2005](#)), multiview large margin distribution machine (MVLDM) ([Hu et al., 2025](#)), and multi-view twin support vector machine (MvTSVM) ([Xie & Sun, 2015](#)). View-1 represents the original data, while view-2 is constructed by retaining 95% of the variance through principal component analysis to achieve dimensionality reduction. The experiments are carried out using a diverse array of datasets sourced from the UCI and KEEL datasets, ensuring a robust assessment across multiple domains. Furthermore, we specifically evaluate the CPBS-MvBLS model using the benchmark multi-view AwA dataset and MNIST dataset to further validate its efficacy. For the experiments, we utilize a computing environment configured with Windows 11 and Python version 3.11, running on an Intel® Xeon® Gold 6226® CPU, equipped with 128 GB of RAM. This setup provides the necessary computational resources to effectively handle the data processing and model training requirements for the experiments. To ensure a fair and rigorous comparison, all the hyperparameters are fine-tuned using 5-fold cross-validation combined with a grid search strategy. The coupling parameter ρ , regularization parameters c , and c_i (for $i = 1, 2, 3$) are selected from the range $\{10^{-5}, 10^{-4}, \dots, 10^5\}$ and we assume $c_1 = c_2 = c_3 = \rho$. The number of feature groups m_Z is sampled from $\{1, 3, \dots, 21\}$, while the enhancement groups are fixed at $m_H = 1$ for simplicity. The number of nodes per feature group (n_Z), the number of nodes per enhancement group (n_H) are varied within the range $\{5, 10, \dots, 50\}$ and set $n_Z = n_H$. The kernel parameter μ is chosen from the set $\{2^{-5}, 2^{-4}, \dots, 2^5\}$, and the steepness (ω) and shape (τ) parameters are optimized from $\left\{\frac{1}{2}, \frac{5}{8}, \frac{3}{4}, \frac{7}{8}, 1\right\}$. The dataset was split into 70% of the training set and 30% of the testing set. The systematic parameter selection ensures that the models are evaluated under an optimal experimental setup, facilitating a reliable comparison with baseline models.

5.1. Evaluation of the CPBS-BLS model using benchmark UCI and KEEL datasets

To evaluate the efficacy of the proposed CPBS-BLS model, we conduct comprehensive experiments on 29 benchmark datasets from the

UCI and KEEL repositories. Performance assessment is based on two key metrics: classification accuracy and average rank. The detailed comparative results are presented in [Table 2](#). The baseline ELM, BLS, IF-BLS, NF-BLS, F-BLS, IF-TSVM, H-ELM, and edRVFL models achieved average accuracies of 75.87%, 76.64%, 77.02%, 77.78%, 76.74%, 68.22%, 76.55%, and 77.89%, respectively. Their corresponding average standard deviations are 8.14, 7.77, 7.13, 8.60, 7.12, 13.89, 8.84, and 7.21, respectively. In contrast, the proposed CPBS-BLS model demonstrated superior performance by attaining an average accuracy of 81.09% with the lowest average standard deviation of 6.08. A lower standard deviation indicates that the proposed model yields more consistent performance across multiple datasets. From a statistical perspective, reduced variability suggests that the model's predictions are less sensitive to fluctuations in training samples and data distributions, thereby reflecting improved stability and robustness. Consequently, the CPBS-BLS framework demonstrates a stronger generalization capability and maintains reliable performance under varying experimental conditions. This significant improvement highlights the effectiveness of our proposed CPBS-BLS model. To further validate the results, we employed a ranking test, where lower ranks indicate better performance. Given $\mathcal{T}$ models and $\mathcal{D}$ datasets, the average rank for model t is computed as: $\mathcal{R}_t = \frac{1}{\mathcal{D}} \sum_{i=1}^{\mathcal{D}} \mathcal{R}(t, i)$. The baseline BLS, ELM, IF-BLS, NF-BLS, F-BLS, IF-TSVM, H-ELM, and edRVFL models yielded average ranks of 5.91, 4.93, 4.33, 4.47, 6.16, 5.88, 5.29, and 4.72, respectively. Our proposed CPBS-BLS model achieved a significantly better rank of 3.31, further validating the competitive superiority of the proposed CPBS-BLS model. The analysis confirms that our proposed CPBS-BLS model is a top-performing model in terms of ranking. To quantify the statistical significance of these results, we conduct four statistical tests: the Friedman test, the Nemenyi post hoc test, the Win-tie-loss test, and the Wilcoxon signed rank test. The Friedman test ([Demšar, 2006](#)) is a non-parametric statistical method for comparing multiple models across different datasets. The chi-squared value for Friedman test statistic is calculated as: $\chi_r^2 = \frac{12\mathcal{D}}{\mathcal{T}(\mathcal{T}+1)} \left[\sum_{t=1}^{\mathcal{T}} \mathcal{R}_t^2 - \frac{\mathcal{T}(\mathcal{T}+1)^2}{4} \right]$ and the corresponding Friedman statistic F_F ([Iman & Davenport, 1980](#)) is then given by: $F_F = \frac{(\mathcal{D}-1)\chi_r^2}{\mathcal{D}(\mathcal{T}-1)-\chi_r^2}$, where $\mathcal{D}$ is the number of datasets, $\mathcal{T}$ is the number of models, and $\mathcal{R}_t$ is the average rank for model t . Under the null hypothesis that all models perform equally, the test statistic follows a χ_r^2 distribution with $(\mathcal{T}-1)$ and $(\mathcal{T}-1)(\mathcal{D}-1)$ degrees of freedom (dof) at $\alpha = 0.05$. We reject the null hypothesis if the computed F_F exceeds the critical value of Friedman's statistic, concluding that there are statistically significant differences among the models' performances. In the case of our experiment, we have $\mathcal{T} = 9$ and $\mathcal{D} = 29$, thus we computed that $\chi_r^2 = 24.3$, Friedman statistic is $F_F = 3.28$, and the critical value of Friedman statistic is 1.98 for dof 8 and 224. Since $1.98 < 3.28$, we reject the null hypothesis and establish a significant statistical difference between the baseline and the proposed CPBS-BLS model. Following a significant Friedman test result, the Nemenyi post-hoc test ([Demšar, 2006](#)) is employed to identify specific pairwise differences between models.

The test computes the critical difference $CD = q_\alpha \sqrt{\frac{\mathcal{T}(\mathcal{T}+1)}{6\mathcal{D}}}$, where CD determines whether performance differences between models are statistically significant at confidence level α , with $\mathcal{T}$ representing the number of models and $\mathcal{D}$ the number of datasets. If the absolute difference between the average ranks of any two models is greater than CD , we reject the null hypothesis of equal performance at significance level α , concluding that the models exhibit statistically significant differences in their overall performance across the evaluated datasets. Given $\mathcal{T} = 9$ and $\mathcal{D} = 29$, the CD is 2.23. The average rank differences between the proposed CPBS-BLS and the baseline models BLS, ELM, IF-BLS, NF-BLS, F-BLS, IF-TSVM, H-ELM and edRVFL are 2.60, 1.62, 1.02, 1.16, 2.84, 2.57, 1.98, and 1.41, respectively. Thereby, the proposed CPBS-BLS model has obtained a significant difference from the BLS, F-BLS, and IF-TSVM based on the Nemenyi post-hoc test. While CPBS-BLS shows no statistically significant differences from ELM, IF-BLS, NF-BLS, H-ELM, and

Table 2

Performance comparison of baseline and proposed CPBS-BLS model on UCI and KEEL datasets in terms of accuracy and standard deviation .

Dataset ↓ Model →	ELM (Huang et al., 2006)	BLS (Chen & Liu, 2017)	IF-BLS (Sajid et al., 2024)	NF-BLS (Feng & Chen, 2018)	F-BLS (Sajid et al., 2024)	IF-TSVM (Rezvani et al., 2019b)	H-ELM (Tang et al., 2015)	edRVFL (Shi et al., 2021)	CPBS-BLS ^a
Best hyperparameters	Acc. ± Std. (c, n_H)	Acc. ± Std. (c, n_Z, m_Z, n_H)	Acc. ± Std. (c, n_Z, m_Z, n_H, μ)	Acc. ± Std. (c, m_Z, n_Z, n_H)	Acc. ± Std. (c, m_Z, n_Z, n_H)	Acc. ± Std. (c, c, μ)	Acc. ± Std. (c, n_H)	Acc. ± Std. (c, n_Z, m_Z, n_H)	Acc. ± Std. ($c, n_Z, m_Z, n_H, \mu, \tau$)
aus	87.98 ± 1.5 (10 ⁻² , 163)	81.99 ± 2.85 (10 ⁻¹ , 15, 17, 15)	87.44 ± 0.96 (10, 15, 1, 15, 4)	78.43 ± 1.22 (1, 30, 3, 95)	75.38 ± 1.52 (10 ⁻³ , 35, 15, 105)	89.9 ± 0.8 (10 ³ , 10 ⁵ , 0.03125)	74.36 ± 1.62 (10 ⁻² , 175)	86.66 ± 2.34 (10 ⁻³ , 2, 538, 10)	89.9 ± 4.65 (10 ⁻³ , 30, 3, 30, 16, 0.5)
bank	89.54 ± 0.57 (10 ³ , 163)	87.91 ± 0.63 (1, 5, 11, 5)	89.41 ± 0.69 (10 ⁵ , 0.5, 45, 19, 15)	89.58 ± 0.7 (10 ⁻⁵ , 15, 15, 45)	88.76 ± 0.65 (10 ⁵ , 15, 15, 65)	89.5 ± 0.56 (10 ⁻⁵ , 10 ⁵ , 1)	88.63 ± 0.59 (10 ⁻¹ , 25)	89.89 ± 7.92 (10 ⁻³ , 1, 538, 3)	90.64 ± 1.51 (10 ⁴ , 5, 1, 5, 16, 0.625)
blood	77.33 ± 13.42 (100, 23)	76.69 ± 13.52 (1, 20, 17, 20)	77.9 ± 13.23 (1, 40, 9, 40, 16)	76.91 ± 13.22 (10 ⁻³ , 5, 7, 5)	75.72 ± 11.45 (100, 15, 7, 65)	76.89 ± 14.98 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	77.17 ± 13.06 (10 ⁻³ , 45)	76.2 ± 4.51 (10 ⁻⁵ , 4, 282, 10)	76.89 ± 5.58 (0.1, 5, 15, 5, 8, 0.625)
breast_cancer	69.77 ± 34.11 (1, 83)	68.6 ± 24.95 (10, 10, 1, 10)	67.79 ± 26.88 (10 ³ , 5, 19, 5, 32)	70.18 ± 44.62 (100, 5, 1, 5)	72.29 ± 27.59 (10 ⁻⁵ , 40, 1, 55)	78.12 ± 44.62 (10 ⁻³ , 10 ⁻⁵ , 0.03125)	84.27 ± 20.78 (10 ⁻³ , 35)	70.13 ± 1.77 (10 ⁻⁵ , 1, 282, 10)	74.42 ± 12.54 (10 ⁻² , 30, 5, 30, 32, 0.75)
breast_cancer_wisc	98.1 ± 7.74 (10 ⁻¹ , 63)	95.1 ± 3.97 (1, 15, 13, 15)	96.1 ± 5.35 (10 ⁵ , 1, 20, 21, 85)	90.71 ± 5.24 (1, 30, 17, 55)	90.23 ± 5.62 (1, 35, 13, 75)	86.23 ± 9.68 (10 ⁻⁵ , 10 ³ , 0.03125)	85.42 ± 6.62 (1, 45)	90.56 ± 2.63 (10 ⁻¹ , 1, 1127, 3)	97.62 ± 3.15 (1, 30, 21, 30, 0.125, 0.5)
breast_cancer_wisc_diag	99.42 ± 2.49 (10 ⁻² , 163)	93.57 ± 3.66 (10, 25, 19, 25)	92.85 ± 1.72 (1, 50, 13, 50, 32)	98.28 ± 5.18 (10 ⁻⁵ , 5, 17, 5)	83.67 ± 3.35 (10 ³ , 40, 3, 5)	69.39 ± 12.61 (10 ⁻⁵ , 10 ⁵ , 0.03125)	89.81 ± 3.41 (10 ⁻¹ , 205)	94.72 ± 5.48 (10 ⁻¹ , 1, 551, 2)	93.47 ± 15.31 (10 ⁻¹ , 20, 21, 20, 0.25, 1)
breast_cancer_wisc_prog	68.33 ± 4.62 (10 ⁻¹ , 83)	78.33 ± 6.27 (10 ⁻⁴ , 45, 1, 45)	97.89 ± 6.28 (10 ⁴ , 4, 45, 21, 15)	79.12 ± 8.77 (1, 5, 19, 55)	83.21 ± 3.62 (100, 25, 1, 105)	79.9 ± 8.94 (10 ⁻⁵ , 10 ⁻³ , 0.03125)	93.2 ± 9.55 (1, 145)	82.36 ± 8.96 (10 ⁻¹ , 0.25, 244, 2)	98.69 ± 8.01 (0.1, 35, 9, 35, 8, 0.875)
chess_krvkp	95.62 ± 8.79 (10 ⁴ , 203)	93.74 ± 2.4 (0.1, 10, 15, 10)	89.73 ± 2.43 (10 ⁵ , 0.03125, 30, 3, 75)	70.4 ± 3.64 (10 ⁻⁵ , 40, 21, 15)	84.29 ± 2.62 (10 ⁻¹ , 40, 5, 15)	25.1 ± 20.55 (10 ⁻⁵ , 10 ³ , 0.03125)	82.1 ± 12.38 (10 ⁻⁵ , 75)	85.26 ± 14.72 (10, 1, 461, 2)	98.63 ± 12.01 (10, 5, 3, 5, 32, 0.75)
congressional_voting	61.07 ± 2.08 (10 ⁻² , 63)	61.39 ± 3.68 (10 ⁴ , 5, 3, 5)	73.89 ± 4.25 (10 ² , 50, 9, 50, 32)	63.91 ± 3.78 (10 ⁻¹ , 5, 3, 25)	60.23 ± 5.78 (1, 30, 9, 85)	60.25 ± 1.74 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	63.24 ± 1.92 (10 ⁻¹ , 95)	62.76 ± 1.85 (10 ⁻¹ , 4, 461, 10)	76.25 ± 4.67 (10 ⁻³ , 45, 17, 45, 16, 0.625)
cylinder_bands	76.62 ± 3.66 (100, 143)	70.13 ± 3.44 (1, 45, 5, 45)	72.85 ± 1.55 (100, 4, 5, 19, 65)	69.53 ± 6.9 (10 ⁻¹ , 15, 13, 95)	69.13 ± 3.28 (1, 50, 13, 85)	60.87 ± 17.94 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	63.24 ± 10.78 (10 ⁻¹ , 55)	66.99 ± 6.41 (10 ⁻⁴ , 1, 1127, 9)	66.23 ± 5.45 (10 ⁴ , 30, 7, 30, 2, 0.5)
echocardiogram	82.5 ± 5.81 (10 ⁻¹ , 103)	80 ± 7.46 (10, 40, 19, 40)	79.55 ± 6.13 (10 ⁻⁴ , 5, 15, 5, 0.25)	73.49 ± 3.65 (1, 15, 3, 105)	69.66 ± 7.27 (100, 10, 1, 105)	83.29 ± 7.01 (10 ⁻⁵ , 10 ⁵ , 0.03125)	74.15 ± 6.17 (10 ⁻¹ , 125)	84.8 ± 2.57 (10 ⁻⁵ , 1, 1076, 5)	79.35 ± 12.3 (10 ⁻³ , 20, 11, 20, 0.0625, 1)
haberman_survival	76.09 ± 8.47 (10 ⁻² , 143)	78.26 ± 7.37 (10 ⁻³ , 10, 9, 10)	79.35 ± 8.51 (10 ⁻⁴ , 5, 7, 5, 4)	70.76 ± 8.48 (1, 5, 1, 5)	78.94 ± 10.06 (10 ⁵ , 45, 3, 5)	82 ± 8.47 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	93.56 ± 8.06 (10 ⁻³ , 55)	73.51 ± 3.19 (10 ⁻⁵ , 4, 282, 10)	77.53 ± 8.78 (10 ⁻⁴ , 5, 13, 5, 32, 0.75)
heart_hungarian	77.53 ± 10.81 (10 ⁻¹ , 83)	84.12 ± 9.71 (10 ⁻² , 25, 1, 25)	83.96 ± 12.55 (10 ³ , 10, 1, 10, 0.125)	87.74 ± 16.37 (10 ⁻⁵ , 10, 1, 15)	84.52 ± 9.58 (10 ⁻³ , 5, 5, 25)	79.35 ± 48.49 (10 ⁻⁵ , 10 ⁵ , 0.03125)	82.58 ± 7.98 (10 ⁻⁵ , 65)	62.81 ± 1.04 (10 ⁻⁵ , 4, 461, 3)	80.85 ± 3.61 (0.1, 30, 19, 30, 8, 0.75)
hepatitis	85.11 ± 9.51 (10 ⁻¹ , 183)	80.85 ± 9.83 (10 ⁻³ , 15, 13, 15)	72.38 ± 4.21 (100, 15, 19, 15, 1)	72.38 ± 5.77 (10 ⁻³ , 45, 17, 55)	59.56 ± 9.51 (1, 40, 19, 35)	71.35 ± 14.17 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	72.39 ± 11.76 (10 ⁻¹ , 105)	84.48 ± 7.66 (100, 1, 263, 8)	70.86 ± 3.67 (10 ⁻⁴ , 5, 1, 5, 0.03125, 0.5)
ilpd_indian_liver	68.57 ± 4.9 (1, 83)	70.29 ± 5.25 (10 ² , 30, 7, 30)	73.28 ± 6.39 (10 ⁻⁴ , 10, 15, 50, 32)	77.83 ± 5.75 (1, 30, 1, 35)	68.69 ± 4.22 (100, 5, 1, 75)	78.11 ± 5.08 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	74.52 ± 4.77 (10 ⁵ , 5)	71.35 ± 2.31 (10 ⁻⁵ , 4, 282, 10)	83.04 ± 3.73 (10 ⁻⁴ , 10, 21, 10, 32, 1)
ionosphere	85.55 ± 8.35 (10 ⁻¹ , 163)	85.85 ± 9.54 (10 ⁻² , 25, 19, 25)	89.2 ± 6.47 (10 ³ , 16, 10, 19, 65)	91.46 ± 3.13 (1, 5, 21, 5)	88.57 ± 8.78 (1, 30, 19, 15)	91.22 ± 21.57 (10 ³ , 10 ⁻⁵ , 0.03125)	88.55 ± 8.42 (1, 175)	91.75 ± 4.92 (10 ⁻¹ , 1, 1127, 5)	85.85 ± 16.97 (10 ⁻⁴ , 5, 1, 5, 4, 0.75)
mammographic	80.28 ± 4.02 (10 ⁴ , 63)	78.89 ± 2.48 (10 ⁻⁴ , 20, 7, 20)	79.55 ± 5.39 (10 ⁵ , 0.25, 10, 19, 85)	79.5 ± 5.59 (1, 15, 13, 15)	78.67 ± 4.88 (10 ³ , 30, 1, 105)	82.3 ± 2.79 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	82.32 ± 2.92 (10 ⁻³ , 5)	78.98 ± 3.88 (10 ⁻³ , 1, 525, 3)	83.04 ± 4.2 (10, 10, 3, 10, 1, 0.625)
monks_1	77.84 ± 8.43 (10, 83)	68.66 ± 11.78 (10, 15, 17, 15)	39.51 ± 9.33 (10 ⁴ , 15, 21, 15, 1)	86.13 ± 10.79 (1, 50, 9, 75)	77.14 ± 7.64 (10 ⁵ , 10, 1, 95)	40.64 ± 5.6 (10 ⁻³ , 10 ⁻³ , 0.03125)	66.22 ± 14.71 (10 ⁻¹ , 135)	89.2 ± 12.73 (10 ⁻¹ , 0.25, 551, 4)	76.65 ± 8.3 (1, 30, 3, 30, 8, 1)
monks_2	64.64 ± 6.51 (1, 203)	53.93 ± 4.78 (100, 15, 1, 15)	67.28 ± 8.15 (10 ³ , 40, 17, 25, 32)	84.34 ± 8.48 (10 ⁻¹ , 40, 13, 95)	68.98 ± 5.1 (10 ⁵ , 45, 19, 105)	65.73 ± 4.58 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	66.23 ± 4.26 (10 ⁻¹ , 205)	87.33 ± 3.54 (1, 1, 512, 3)	72.35 ± 3.42 (0.1, 10, 5, 10, 8, 0.5)
musk_1	82.52 ± 4.84 (10 ⁻² , 163)	81.82 ± 11.85 (0.1, 10, 21, 10)	78.2 ± 7.75 (10 ³ , 16, 10, 17, 65)	67.85 ± 8.8 (10 ⁻¹ , 35, 21, 5)	76.68 ± 7.25 (10 ⁻¹ , 30, 13, 5)	22.11 ± 36.05 (10 ³ , 10 ⁻⁵ , 0.03125)	89.49 ± 6.28 (10 ⁻³ , 175)	71.84 ± 2.11 (10 ² , 2, 1076, 7)	79.02 ± 5.18 (10 ⁻³ , 30, 13, 30, 16, 1)

Table 2

Continue.

Dataset ↓ Model →	ELM (Huang et al., 2006)	BLS (Chen & Liu, 2017)	IF-BLS (Sajid et al., 2024)	NF-BLS (Feng & Chen, 2018)	F-BLS (Sajid et al., 2024)	IF-TSVM (Rezvani et al., 2019b)	H-ELM (Tang et al., 2015)	edRVFL (Shi et al., 2021)	CPBS-BLS ^a
pima	58.61 ± 2.29 (0.1, 63)	74.46 ± 2.89 (1, 35, 19, 35)	84.98 ± 1.58 (10 ⁴ , 45, 3, 25, 8)	72.79 ± 4.2 (10 ⁻¹ , 5, 9, 65)	65.1 ± 3.82 (100, 20, 1, 5)	71.86 ± 5.95 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	70.31 ± 4.99 (1, 55)	73.69 ± 19.06 (10 ⁻¹ , 1, 256, 10)	78.17 ± 10.85 (100, 20, 9, 20, 16, 0.75)
pittsburg_bridges_T_OR_D	70.28 ± 11.56 (10 ⁻¹ , 83)	83.87 ± 5.42 (10 ³ , 5, 21, 5)	90.24 ± 5.99 (10 ³ , 2, 40, 5, 95)	90.14 ± 7.95 (1, 40, 1, 75)	88.19 ± 8.35 (100, 30, 3, 15)	86.14 ± 13.94 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	87.29 ± 6.52 (10 ⁻³ , 125)	86.15 ± 2.84 (10 ⁻⁵ , 4, 282, 10)	87.1 ± 0.46 (1, 35, 5, 35, 8, 0.75)
planning	41.82 ± 8.85 (10 ⁻¹ , 23)	70.24 ± 9.21 (10 ³ , 20, 11, 20)	74.93 ± 8.83 (10 ³ , 16, 10, 17, 65)	72.49 ± 9.53 (10 ⁻³ , 10, 1, 105)	71.38 ± 9.58 (100, 5, 1, 45)	71.38 ± 8.85 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	71.38 ± 8.85 (10 ⁻³ , 5)	73.01 ± 2.56 (10 ⁻⁵ , 1, 250, 9)	76.36 ± 6.83 (10 ⁴ , 5, 1, 5, 16, 0.5)
shuttle_6_vs_2-3	45 ± 25.59 (0.1, 63)	100 ± 28.36 (10, 40, 15, 40)	44.93 ± 22.33 (10 ⁻⁴ , 5, 1, 35, 0.03125)	56.38 ± 14.57 (10 ⁻¹ , 5, 19, 55)	54.34 ± 15.63 (1, 30, 19, 105)	100 ± 0 (10 ⁻⁵ , 10 ⁵ , 0.03125)	43.98 ± 21.02 (1, 205)	63.87 ± 11.77 (10 ⁻³ , 0.25, 231, 1)	100 ± 0 (1, 20, 13, 20, 16, 1)
spect	55 ± 3.52 (10 ³ , 23)	66.79 ± 8.37 (10 ⁻¹ , 25, 3, 25)	67.5 ± 7.38 (0.1, 15, 13, 15, 32)	67.92 ± 6.93 (10 ⁻¹ , 45, 11, 55)	65 ± 2.79 (1, 35, 5, 95)	58.49 ± 14.05 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	68.75 ± 6.61 (1, 35)	69.82 ± 8.44 (10 ⁻¹ , 10, 256, 10)	69.43 ± 10.91 (100, 30, 17, 30, 16, 0.875)
spectf	83.95 ± 20.89 (1, 23)	66.79 ± 20.89 (10, 5, 1, 5)	73.9 ± 14.67 (100, 16, 25, 9, 25)	67.92 ± 18.9 (10 ⁻¹ , 10, 1, 65)	65 ± 20.44 (100, 5, 1, 105)	58.49 ± 20.89 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	68.75 ± 20.89 (10 ⁻³ , 5)	79.39 ± 23.89 (10 ⁻⁵ , 4, 282, 10)	69.43 ± 4.7 (10, 45, 9, 45, 4, 0.75)
statlog_australian_credit	70.67 ± 1.14 (10 ⁻⁴ , 3)	53.85 ± 1.48 (10 ³ , 5, 17, 5)	66.61 ± 1.88 (10 ² , 5, 1, 5, 16)	69.28 ± 3.14 (1, 5, 17, 5)	67.83 ± 1.5 (100, 5, 1, 95)	70.67 ± 1.74 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	67.83 ± 1.74 (10 ⁻⁵ , 5)	67.82 ± 8.66 (10 ⁻⁵ , 4, 282, 10)	86.24 ± 4.64 (100, 5, 1, 5, 8, 0.875)
statlog_heart	82.72 ± 3.56 (10 ⁻⁴ , 63)	69.08 ± 2.11 (10 ⁻² , 5, 5, 5)	85.19 ± 4.83 (10 ³ , 15, 15, 15, 0.125)	88.74 ± 1.01 (10 ⁻² , 40, 15, 45)	82.22 ± 2.8 (1, 50, 7, 105)	55.56 ± 2.61 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	80.74 ± 3.84 (10 ⁻¹ , 55)	80 ± 2.14 (10 ⁻¹ , 2, 256, 10)	86.42 ± 6.83 (10 ⁻³ , 25, 19, 25, 4, 0.5)
tic_tac_toe	99.65 ± 7.91 (10, 143)	69.2 ± 1.28 (0.1, 50, 15, 50)	79.35 ± 0.88 (10 ² , 20, 19, 20, 16)	82.76 ± 13.13 (10 ⁻¹ , 35, 15, 95)	97.81 ± 1.7 (1, 15, 21, 45)	65.31 ± 48.41 (10 ⁻⁵ , 10 ⁻⁵ , 0.03125)	75.73 ± 25.87 (10 ⁻⁵ , 85)	63.52 ± 29.27 (10 ⁻¹ , 1, 10 ³ , 4)	99.65 ± 0 (10 ³ , 15, 15, 15, 16, 0.75)
Average accuracy	75.87 ± 8.14	76.64 ± 7.77	77.02 ± 7.13	77.78 ± 8.60	76.74 ± 7.12	68.22 ± 13.89	76.55 ± 8.84	77.89 ± 7.21	81.09 ± 6.08

^a Represents the proposed model. Boldface highlights the model with the highest average accuracy. Avg. and Std. are acronyms of average and standard deviation.

Table 3

Win-tie-loss test on UCI and KEEL datasets for the proposed CPBS-BLS model.

	BLS (Chen & Liu, 2017)	ELM (Huang et al., 2006)	IF-BLS (Sajid et al., 2024)	NF-BLS (Feng & Chen, 2018)	F-BLS (Sajid et al., 2024)	IF-TSVM (Rezvani et al., 2019b)	H-ELM (Tang et al., 2015)	edRVFL (Shi et al., 2021)	CPBS-BLS ^a
ELM (Huang et al., 2006)	[19, 0, 10]								
IF-BLS (Sajid et al., 2024)	[20, 0, 9]	[12, 0, 17]							
NF-BLS (Feng & Chen, 2018)	[19, 0, 10]	[15, 0, 14]	[14, 0, 15]						
F-BLS (Sajid et al., 2024)	[12, 0, 17]	[12, 0, 17]	[8, 0, 21]	[6, 0, 23]					
IF-TSVM (Rezvani et al., 2019b)	[12, 1, 16]	[13, 1, 15]	[11, 0, 18]	[5, 2, 22]	[12, 0, 17]				
H-ELM (Tang et al., 2015)	[16, 0, 13]	[12, 0, 17]	[11, 0, 18]	[11, 0, 18]	[12, 2, 15]	[21, 2, 6]			
edRVFL (Shi et al., 2021)	[19, 0, 10]	[14, 0, 15]	[12, 0, 17]	[16, 0, 13]	[19, 0, 10]	[19, 0, 10]	[16, 0, 13]		
CPBS-BLS ^a	[22, 1, 6]	[17, 3, 9]	[21, 0, 8]	[17, 0, 12]	[22, 0, 7]	[24, 1, 4]	[19, 0, 10]	[20, 0, 9]	

^a Represents the proposed model.

edRVFL in the post hoc test, CPBS-BLS is still superior to all baselines by getting both the greatest mean accuracy and the smallest average rank.

The Win-tie-loss sign test (Demšar, 2006) compares multiple models across datasets by tracking pairwise performance outcomes. Under the null hypothesis, each model should win on $\mathcal{D}/2$ datasets. A model is statistically superior if the number of wins exceeds the threshold $= \frac{\mathcal{D}}{2} + \frac{1.96\sqrt{\mathcal{D}}}{2}$ value. For $\mathcal{D} = 29$ datasets, models require ≥ 20 wins (rounded up from 19.77) to demonstrate statistical significance. Furthermore, the ties are evenly distributed between the compared models when the number of ties are even, and one of the tie is disregarded when

the number of ties are odd. Table 3 shows the wins, ties, and losses of the models in the first column against models in the first row. Upon observing Table 3, the proposed model CPBS-BLS wins over BLS, IF-BLS, F-BLS, IF-TSVM, and edRVFL models by winning 22, 21, 22, 24, and 20 datasets. Although the proposed model could not demonstrate a statistical difference from the ELM, NF-BLS, and H-ELM models, but outperformed all the models in terms of highest accuracy, lowest rank, and the greatest number of wins. Moreover, we apply the Wilcoxon signed rank (Demšar, 2006) test to examine whether the proposed CPBS-BLS model differs significantly, in a pairwise sense, from the competing baseline methods. Under the null hypothesis, CPBS-BLS and each baseline are

Table 4

Wilcoxon signed-rank test between the proposed CPBS-BLS model and baseline models on UCI and KEEL datasets.

CPBS-BLS (Proposed) ↓ Baseline →	ELM (Huang et al., 2006)	BLS (Chen & Liu, 2017)	IF-BLS (Sajid et al., 2024)	NF-BLS (Feng & Chen, 2018)	F-BLS (Sajid et al., 2024)	IF-TSVM (Rezvani et al., 2019b)	H-ELM (Tang et al., 2015)	edRVFL (Shi et al., 2021)
p-value	0.023	0.0021	0.084	0.027	1.04E-04	3.26E-04	0.006	0.04
Null Hypothesis	Rejected	Rejected	Not Rejected	Rejected	Rejected	Rejected	Rejected	Rejected

assumed to deliver comparable performance. For each dataset, we first compute the performance difference between CPBS-BLS and the corresponding baseline model; we then rank the absolute values of these differences and assign to each rank the sign of its original difference. When the resulting p-value for a comparison is below 0.05, we reject the null hypothesis and conclude that the method with the lower average rank outperforms the one with the higher rank. Table 4 reports the corresponding p-values together with the decision on the null hypothesis. Overall, the pairwise analyses between CPBS-BLS and the baseline models reveal a clear pattern: the frequent rejection of the null hypothesis in favor of CPBS-BLS provides strong statistical evidence of its superiority.

5.2. Evaluation of the CPBS-MvBLS model using benchmark AWA datasets

This section presents the experimental results obtained from evaluating the performance of the proposed CPBS-MvBLS model on 30 Animal with Attribute (AWA) datasets. Table 5 displays the classification accuracy for all baseline and proposed models. The baseline models ELM1, ELM2, BLS1, BLS2, SVM2K, MVLDM, and MvTSVM achieve average accuracies of 69.91%, 75.93%, 70.81%, 77.61%, 76.63%, 72.7%, and 64.33%, respectively. Their corresponding average standard deviations are 5.3, 5.26, 5.65, 5.34, 5.05, 5.70, and 4.53. In comparison, our proposed CPBS-MvBLS model demonstrates superior performance with an average accuracy of 78.45% with an average standard deviation of 3.99, substantially outperforming all existing frameworks. Furthermore, the CPBS-MvBLS model attains the lowest mean rank of 2.78, while the baseline models ELM1, ELM2, BLS1, BLS2, SVM2K, MVLDM, and MvTSVM obtained considerably higher average ranks of 5.97, 4.05, 5.8, 2.93, 3.1, 4.77, and 6.6, respectively. These outcomes demonstrate that the proposed CPBS-MvBLS model outperforms all of the baseline models.

Statistical analysis using the Friedman test with $\mathcal{T} = 8$ and $\mathcal{D} = 30$ yielded $\chi_r^2 = 88.84$ and Friedman static is $F_F = 21.26$. Given that the critical Friedman value is 2.05 for *dof* 7 and 203. Since $21.26 > 2.05$, we reject the null hypothesis and confirm the statistically significant differences between the performance of the model. The CD for the Nemenyi post-hoc test is calculated as 1.92. The average rank difference between the proposed CPBS-MvBLS and the existing models ELM1, ELM2, BLS1, BLS2, SVM2K, MVLDM, and MvTSVM are 3.91, 1.27, 3.02, 0.15, 0.32, 1.99, and 3.28, respectively. After comparing to the CD value, CPBS-MvBLS exhibits a statistical difference from ELM1, BLS1, MVLDM, and MvTSVM. The remaining baseline ELM2, BLS2, and SVM2K models are outperformed by the proposed model in terms of both average accuracy and average ranking. For the Win-tie-loss sign test, the established threshold is 21 (rounded up from 20.37). As shown in Table 6, CPBS-MvBLS achieved significant differences from ELM1, ELM2, BLS1, MVLDM, and MvTSVM with victories of 27, 22, 25, 23, and 28 wins, respectively. These results further validate the superior performance of our proposed CPBS-MvBLS model. For the Wilcoxon signed rank test, p-values are given in the Table 7. Both the average accuracy and average rank metrics indicate that the proposed model consistently outperforms.

5.3. Evaluation of the CPBS-MvBLS model using benchmark UCI and KEEL datasets

Our comprehensive evaluation of the proposed CPBS-MvBLS framework on standard benchmark datasets from UCI and KEEL repositories

demonstrates its superior classification performance, as evidenced by the comparative accuracy results presented in Table 8. The experimental results demonstrate that CPBS-MvBLS achieves the highest average accuracy of 85.63% while maintaining the lowest average standard deviation of 5.16, indicating both superior predictive performance and strong stability across datasets. It consistently outperforms all competing models, including ELM1, ELM2, BLS1, BLS2, SVM2K, MVLDM and MvTSVM with average accuracies of 80.35%, 77.63%, 81.26%, 83.23%, 82.72%, 82.01%, and 81.30%. Their corresponding average standard deviations are 5.37, 5.48, 7.41, 7.94, 6.59, 5.43, and 7.26. Further analysis of average rankings shows CPBS-MvBLS achieving the top 2.87 rank, whereas ELM1, ELM2, BLS1, BLS2, SVM2K, MVLDM, and MvTSVM achieve 4.75, 5.25, 5.22, 3.82, 4.28, 4.7, and 5.12 ranks, respectively.

To statistically validate these findings, we conduct the Friedman test with $\mathcal{T} = 8$ models and $\mathcal{D} = 30$ datasets, yielding $\chi_r^2 = 82.68$ and $F_F = 18.83$, which significantly exceeds the critical value of 2.05 with *dof* 7 and 203 at $\alpha = 0.05$, leading us to reject the null hypothesis. Nemenyi post hoc analysis with a critical difference of 1.92 confirms that CPBS-MvBLS significantly outperforms ELM2, BLS1, and MvTSVM with rank differences of 2.38, 2.35, and 2.25. Although CPBS-MvBLS did not exhibit statistically significant differences from ELM1, BLS2, SVM2K, and MVLDM, with respective rank differences of 1.88, 0.95, 1.41, and 1.83, it achieved higher average accuracy and ranking compared to these baselines. The Win-tie-loss test (Table 9) with threshold = 21 (round up from 20.367) further substantiates these results, showing CPBS-MvBLS achieving decisive victories against ELM1 (22 wins), ELM2 (22 wins), BLS1 (23 wins), MVLDM (21 wins), and MvTSVM (23 wins) while maintaining consistent superiority over all other baseline models across both accuracy and ranking metrics. Table 10 summarizes the Wilcoxon signed-rank test results, where the null hypothesis is consistently rejected in the majority of comparisons. This statistical evidence highlights the superior performance and robustness of the proposed CPBS-MvBLS model over existing baseline models.

5.4. Evaluation of the CPBS-MvBLS model using benchmark MNIST datasets

To assess the performance of the proposed CPBS-MvBLS model on larger-scale and more complex datasets, we further evaluated the proposed model on the benchmark MNIST dataset. MNIST consists of grayscale images of handwritten digits (10 classes) and is widely used to assess the scalability and robustness of classification algorithms on high-dimensional input spaces. To align MNIST with the binary classification setting of CPBS-MvBLS, the original ten-class dataset was reorganized into seven pairwise subsets: 0vs1, 1vs2, 2vs3, 2vs5, 2vs7, 3vs4, and 5vs6. Each subset includes only samples from the corresponding two digits, while retaining the original balanced class distribution. In this experiment, each sample is represented by its raw pixel intensities, and CPBS-MvBLS is applied in a multi-view setting by constructing complementary feature spaces derived from the original images using PCA feature extraction. We compared CPBS-MvBLS with several representative baseline methods under the same experimental setup. For all methods, hyperparameters are tuned using cross-validation, and the best-performing configuration for each model is reported. Table 11 summarizes the classification accuracy and corresponding optimal hyperparameters for the proposed CPBS-MvBLS model and the baseline models. As shown in the table, CPBS-MvBLS achieves competitive or superior

Table 5

Performance comparison of baseline and proposed CPBS-MvBLS model on AwA datasets in terms of accuracy and standard deviation.

Dataset ↓ Model →	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)	CPBS-MvBLS ^a
	Acc. ± Std. (c_1, n_H)	Acc. ± Std. (c_1, n_H)	Acc. ± Std. (c_1, n_Z, m_Z, n_H)	Acc. ± Std. (c_1, n_Z, m_Z, n_H)	Acc. ± Std. (c_1, μ)	Acc. ± Std. (c_1, c_2, c_3, μ)	Acc. ± Std. (c_1, c_2, μ)	Acc. ± Std. (c_1, n_Z, m_Z, n_H, τ)
Chimpanzee vs Giantpanda	71.53 ± 7.4 (10^{-3} , 123)	72.92 ± 6.34 (10^{-4} , 163)	72.92 ± 8.19 (10^{-3} , 40, 15, 40)	82.64 ± 7.03 (10^{-1} , 40, 21, 40)	84.03 ± 3.32 (2, 1)	72.22 ± 5.22 (10^2 , 10^{-3} , 10^{-1} , 4)	47.22 ± 2.5 (10^{-4} , 10^{-4} , 10^{-4})	82.64 ± 4.64 (1, 40, 15, 40, 0.0625)
Giantpanda vs Leopard	61.11 ± 9.49 (1, 43)	78.47 ± 2.91 (10^{-4} , 203)	65.97 ± 4.88 (10^{-1} , 35, 19, 35)	83.33 ± 5.17 (10^{-1} , 35, 19, 35)	80.19 ± 7.62 (4, 1)	61.81 ± 4 (10^2 , 10^{-4} , 10^{-1} , 0.25)	54.17 ± 5.06 (10^{-4} , 10^{-4} , 10^{-4})	82.64 ± 3.62 (1, 50, 11, 50, 0.5)
Giantpanda vs Persiancat	71.53 ± 1.18 (10^{-4} , 183)	76.39 ± 3.74 (10^{-4} , 183)	77.78 ± 9.54 (10^{-2} , 30, 5, 30)	81.25 ± 4.22 (10^{-1} , 50, 19, 50)	81.81 ± 5.51 (2, 0.5)	66.67 ± 1.36 (10^4 , 10^{-2} , 100, 2)	52.08 ± 4.23 (10^{-4} , 10^{-4} , 10^{-4})	84.72 ± 4.38 (10, 20, 15, 20, 1)
Giantpandavs-Raccoon	68.06 ± 7.26 (10^{-3} , 203)	74.31 ± 3.63 (10^{-4} , 163)	68.06 ± 5.53 (10^{-2} , 35, 13, 35)	78.47 ± 7.85 (10^{-2} , 30, 19, 30)	80.19 ± 5.22 (4, 8)	64.58 ± 4.01 (10^4 , 10^{-4} , 10^{-4} , 2)	52.78 ± 3.54 (10^{-4} , 10^{-4} , 10^{-4})	74.31 ± 2.34 (1, 50, 13, 50, 0.25)
Giantpanda vs Seal	80.56 ± 3.92 (10^{-3} , 203)	77.08 ± 6.57 (10^{-4} , 123)	78.47 ± 5.3 (10^{-2} , 20, 21, 20)	81.25 ± 4.66 (10^{-1} , 50, 19, 50)	85.89 ± 6.52 (2, 1)	86.81 ± 5.66 (10^4 , 10^{-2} , 10^{-1} , 2)	56.94 ± 6.23 (10^{-4} , 10^{-4} , 10^{-4})	77.78 ± 2.4 (0.1, 45, 19, 45, 0.5)
Leopard vs Persiancat	70.83 ± 4.78 (10^{-4} , 143)	84.72 ± 4.33 (10^{-4} , 203)	74.31 ± 5.97 (10^{-4} , 50, 21, 50)	86.11 ± 3.34 (10^{-4} , 50, 21, 50)	82.19 ± 5.62 (8, 8)	80.56 ± 6.36 (10^{-1} , 10^{-4} , 0.1, 4)	79.31 ± 1.95 (10^{-4} , 10^{-4} , 10^{-4})	85.42 ± 3.34 (1, 30, 21, 30, 0.25)
Leopard vs Pig	68.75 ± 4.06 (100, 23)	75 ± 7.47 (10^{-4} , 163)	61.11 ± 4.45 (10^{-1} , 15, 13, 15)	72.92 ± 7.68 (10^{-1} , 30, 15, 30)	75 ± 2.83 (2, 0.5)	68.75 ± 7.31 (10^4 , 10^{-4} , 10^{-2} , 2)	61.39 ± 2.24 (10^{-4} , 10^{-4} , 10^4)	79.17 ± 2.37 (1, 20, 19, 20, 4)
Chimpanzee vs Leopard	63.89 ± 7.21 (1, 43)	83.33 ± 4.69 (10^{-4} , 163)	75 ± 5.86 (10^{-1} , 15, 13, 15)	81.94 ± 8.25 (10^{-1} , 50, 19, 50)	80.11 ± 7.99 (1, 0.5)	68.75 ± 5.17 (10^4 , 10^{-2} , 10^{-2} , 4)	46.53 ± 5.24 (10^{-4} , 10^{-4} , 10^{-4})	78.47 ± 4.75 (10^{-4} , 40, 7, 40, 1)
Leopard vs Hippopotamus	63.89 ± 4.8 (10^{-3} , 183)	77.08 ± 2.93 (10^{-4} , 163)	73.61 ± 6.02 (0.1, 5, 19, 5)	77.08 ± 5.89 (10^{-1} , 50, 15, 50)	78.17 ± 3.2 (1, 1)	75 ± 7.71 (100, 10^{-4} , 10^{-1} , 4)	50.69 ± 1.34 (10^{-4} , 10^{-4} , 10^{-4})	70.14 ± 2.24 (10^3 , 20, 17, 20, 16)
Leopard vs Humpbackwhale	89.58 ± 4.02 (1, 63)	91.67 ± 4.42 (10^{-4} , 203)	90.28 ± 2.5 (10^{-1} , 50, 5, 50)	95.83 ± 2.57 (10^{-1} , 50, 13, 50)	90.75 ± 2.7 (4, 0.5)	89.58 ± 5.3 (10^4 , 10^{-2} , 10^{-4} , 0.25)	79.31 ± 5.09 (10^{-4} , 10^{-4} , 10^{-4})	90.97 ± 4.13 (1, 15, 13, 15, 0.03125)
Leopard vs Rat	72.22 ± 5.57 (10^{-3} , 183)	76.39 ± 6.53 (10^{-4} , 83)	62.5 ± 5.94 (10^{-4} , 50, 1, 50)	72.92 ± 5.53 (10^{-1} , 35, 13, 35)	76.42 ± 5.59 (1, 0.5)	65.28 ± 5.03 (10^3 , 10^{-4} , 10^{-1} , 4)	68.61 ± 1.34 (10^{-4} , 10^{-4} , 10^{-4})	73.61 ± 4.92 (1, 45, 9, 45, 8)
Persiancat vs Pig	63.89 ± 2.63 (10^2 , 63)	67.36 ± 5.77 (10^{-3} , 203)	61.81 ± 8.2 (10^{-1} , 35, 9, 35)	78.47 ± 6.4 (10^{-1} , 35, 21, 35)	70 ± 7.69 (1, 2)	69.44 ± 4.85 (10^2 , 10^{-4} , 10^{-1} , 0.25)	69.31 ± 6.67 (10^{-4} , 10^{-4} , 10^{-4})	71.53 ± 2.87 (1, 50, 5, 50, 0.03125)
Persiancat vs Hippopotamus	75.69 ± 3.95 (10^{-4} , 203)	79.86 ± 3.04 (10^{-4} , 203)	70.14 ± 4.55 (0.1, 10, 7, 10)	81.94 ± 5.16 (10^{-1} , 30, 19, 30)	76.81 ± 4.22 (2, 1)	75.69 ± 1.36 (10^4 , 10^{-2} , 10^{-3} , 0.25)	76.53 ± 5.43 (10^{-4} , 10^{-4} , 10^{-4})	80.56 ± 4.36 (1, 20, 21, 20, 0.125)
Persiancat vs Humpbackwhale	70.84 ± 5.04 (10^{-3} , 43)	88.19 ± 2.71 (10^{-4} , 183)	81.25 ± 5.61 (10^{-3} , 15, 13, 15)	93.06 ± 6.65 (10^{-1} , 50, 21, 50)	71.67 ± 6.36 (1, 1)	85.42 ± 4.19 (10^{-3} , 10^{-4} , 10^{-3} , 4)	71.39 ± 4.07 (10^{-4} , 10^{-4} , 10^{-4})	91.67 ± 6.38 (10^{-4} , 35, 19, 35, 8)
Persiancat vs Raccoon	65.28 ± 5.75 (10^{-4} , 163)	81.25 ± 3.33 (10^{-4} , 183)	73.61 ± 8.37 (10^{-1} , 45, 11, 45)	83.33 ± 0.7 (10^{-2} , 45, 15, 45)	82.64 ± 4.45 (4, 1)	65.97 ± 2.32 (10^{-2} , 10^{-3} , 10^3 , 0.25)	79.31 ± 2.57 (10^{-4} , 10^{-4} , 10^{-4})	81.94 ± 5.7 (100, 40, 21, 40, 1)
Persiancat vs Seal	72.22 ± 3.02 (10^{-3} , 203)	71.53 ± 6.55 (10^{-4} , 183)	71.53 ± 5.24 (10^{-1} , 35, 21, 35)	84.72 ± 4.7 (10^{-1} , 50, 19, 50)	80.42 ± 4.6 (1, 4)	83.33 ± 7.2 (10^{-4} , 10^{-2} , 1, 4)	73.47 ± 3.65 (10^{-4} , 10^{-4} , 10^4)	81.25 ± 3.5 (0.1, 40, 19, 40, 16)
Pig vs Humpbackwhale	82.64 ± 2.18 (0.1, 63)	89.58 ± 4.89 (10^{-4} , 183)	82.64 ± 1.91 (10^{-1} , 40, 13, 40)	70.83 ± 8.48 (10^{-1} , 45, 19, 45)	80.19 ± 3.47 (8, 16)	88.89 ± 4.86 (10^{-2} , 10^{-1} , 10^{-4} , 4)	77.92 ± 3.35 (10^{-4} , 10^{-4} , 10^{-4})	86.81 ± 1.93 (1, 50, 5, 50, 2)
Pig vs Raccoon	61.11 ± 7.01 (10^{-4} , 83)	72.92 ± 3.46 (10^{-4} , 123)	59.72 ± 7.5 (0.1, 5, 19, 5)	60.42 ± 5.8 (10^{-1} , 50, 21, 50)	71.69 ± 3.96 (8, 32)	62.5 ± 6.28 (10^3 , 10^{-3} , 10^{-1} , 0.25)	69.31 ± 2.9 (10^{-4} , 10^{-4} , 10^4)	73.61 ± 3.86 (10, 35, 3, 35, 8)

Table 5
Continue.

Dataset ↓ Model →	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)	CPBS-MvBLS ^a
Pig vs Rat	62.5 ± 5.21 (10 ⁻³ , 183)	59.72 ± 6.48 (10 ⁻⁴ , 203)	64.58 ± 3.71 (10 ⁻¹ , 40, 17, 40)	50 ± 4.77 (10 ⁻² , 25, 17, 25)	71.53 ± 7.58 (2, 2)	64.58 ± 7.87 (10 ⁴ , 10 ⁻² , 10, 0.25)	68.61 ± 3.17 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	61.11 ± 5.29 (1, 15, 17, 15, 0.03125)
Pig vs Seal	70.14 ± 5.76 (10 ⁻³ , 143)	68.06 ± 8.01 (10 ⁻⁴ , 203)	70.83 ± 9.76 (10 ⁻² , 50, 13, 50)	65.97 ± 5.71 (10 ⁻¹ , 25, 13, 25)	72.69 ± 6.52 (2, 1)	72.92 ± 5.66 (10 ⁴ , 10 ⁻³ , 10 ⁻² , 0.25)	65.56 ± 9.01 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	69.44 ± 2.56 (1, 50, 21, 50, 8)
Hippopotamus vs Raccoon	72.22 ± 4.59 (10 ⁻³ , 183)	70.14 ± 4.82 (10 ⁻³ , 183)	67.36 ± 7.06 (10 ⁻¹ , 25, 9, 25)	79.17 ± 4.97 (10 ⁻¹ , 30, 19, 30)	78.47 ± 3.17 (4, 8)	75.69 ± 6.58 (10 ⁻¹ , 10 ⁻² , 100, 4)	75.14 ± 6.33 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	81.25 ± 7.77 (10, 50, 9, 50, 4)
Hippopotamus vs Rat	65.28 ± 7.97 (10 ⁻³ , 143)	71.53 ± 8.2 (10 ⁻⁴ , 183)	65.97 ± 4 (10 ⁻¹ , 50, 15, 50)	68.75 ± 3.54 (10 ⁻¹ , 25, 15, 25)	75.33 ± 6.4 (1, 0.5)	64.58 ± 4.63 (10 ³ , 10 ⁻³ , 10 ⁻² , 4)	75.83 ± 3.31 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	75 ± 6.22 (1, 50, 7, 50, 0.5)
Chimpanzee vs Pig	68.75 ± 7.74 (10 ⁻³ , 83)	81.25 ± 4.65 (10 ⁻⁴ , 183)	65.97 ± 2.73 (10 ⁻³ , 20, 13, 20)	81.25 ± 6.29 (10 ⁻¹ , 35, 17, 35)	50.42 ± 5.34 (2, 0.5)	66.67 ± 8.12 (10 ³ , 10 ⁻² , 10 ⁻³ , 4)	51.39 ± 4.71 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	77.78 ± 2.75 (1, 40, 7, 40, 0.125)
Raccoon vs Rat	59.72 ± 7.74 (100, 43)	70.83 ± 6.07 (10 ⁻⁴ , 163)	56.25 ± 7.4 (10 ⁻¹ , 15, 11, 15)	68.75 ± 5.06 (10 ⁻⁴ , 25, 17, 25)	62.22 ± 6.3 (2, 0.5)	65.28 ± 7.44 (10 ⁴ , 10 ⁻⁴ , 10, 4)	61.89 ± 7.47 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	67.36 ± 2.11 (1, 40, 9, 40, 0.125)
Raccoon vs Seal	78.47 ± 7.14 (10 ⁻⁴ , 183)	84.03 ± 6.5 (10 ⁻⁴ , 183)	79.17 ± 6.19 (10 ⁻⁴ , 35, 5, 35)	87.5 ± 5.85 (10 ⁻¹ , 45, 21, 45)	90.28 ± 3.29 (1, 2)	75.69 ± 7.87 (10 ⁴ , 10 ⁻⁴ , 10 ⁻¹ , 4)	75.39 ± 4.27 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁴)	87.5 ± 3.8 (10 ⁻³ , 10, 19, 10, 4)
Rat vs Seal	68.75 ± 4.46 (10 ⁻⁴ , 183)	68.75 ± 7.58 (10 ⁻⁴ , 143)	69.44 ± 2.4 (10 ⁻² , 40, 13, 40)	65.28 ± 3.25 (10 ⁻¹ , 40, 19, 40)	70.86 ± 3.63 (2, 1)	69.87 ± 6.56 (10 ⁴ , 10 ⁻² , 10 ⁻¹ , 0.25)	65.17 ± 2.89 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	71.53 ± 4.23 (1, 45, 13, 45, 1)
Chimpanzee vs Hippopotamus	71.53 ± 5.76 (10 ² , 43)	70.14 ± 4.44 (10 ⁻⁴ , 203)	77.08 ± 3.13 (10 ⁻⁴ , 45, 19, 45)	79.86 ± 2.5 (10 ⁻¹ , 15, 11, 15)	70.94 ± 3.85 (2, 32)	78.47 ± 6.64 (10 ⁻² , 10 ⁻⁴ , 10 ⁻⁴ , 4)	54.86 ± 8.66 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	77.08 ± 6.48 (10, 15, 21, 15, 0.0625)
Chimpanzee vs Raccoon	69.44 ± 6.12 (10 ⁻³ , 203)	63.89 ± 7.01 (10 ⁻⁴ , 123)	68.06 ± 3.51 (10 ⁻¹ , 20, 11, 20)	75.69 ± 8.43 (10 ⁻¹ , 50, 21, 50)	80.33 ± 5 (4, 32)	72.22 ± 8.77 (10 ³ , 10 ⁻⁴ , 10 ⁻¹ , 4)	63.47 ± 4.48 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁴)	77.78 ± 4.64 (1, 45, 21, 45, 2)
Chimpanzee vs Rat	57.64 ± 2.75 (10 ⁴ , 43)	75 ± 5.76 (10 ⁻⁴ , 183)	63.89 ± 4.09 (10 ⁻⁴ , 10, 17, 10)	78.47 ± 6.99 (0.1, 35, 13, 35)	77.08 ± 4.32 (1, 4)	68.06 ± 6.91 (10 ⁴ , 10 ⁻³ , 10 ⁻⁴ , 4)	52.78 ± 7.15 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	77.08 ± 3.23 (10 ⁻⁴ , 40, 7, 40, 1)
Chimpanzee vs Seal	79.17 ± 4.49 (10 ⁻³ , 183)	76.39 ± 4.82 (10 ⁻⁴ , 143)	75 ± 4.21 (10 ⁻⁴ , 5, 19, 5)	81.11 ± 2.62 (10 ⁻² , 50, 11, 50)	70.69 ± 5.17 (2, 16)	75.69 ± 5.82 (10 ⁻² , 10 ⁻² , 10 ⁻¹ , 0.25)	53.47 ± 6.97 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	83.33 ± 2.8 (1, 50, 11, 50, 0.5)
Average accuracy	69.91 ± 5.3	75.93 ± 5.26	70.81 ± 5.65	77.61 ± 5.34	76.63 ± 5.05	72.7 ± 5.70	64.33 ± 4.53	78.45 ± 3.99

^a Represents the proposed model. Boldface highlights the model with the highest average accuracy. Avg. and Std. are acronyms of average and standard deviation.

Table 6
Win-tie-loss test on AWA datasets for the proposed CPBS-MvBLS model.

	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)
ELM2 (Huang et al., 2006)	[21, 1, 8]						
BLS1 (Chen & Liu, 2017)	[16, 2, 12]	[7, 2, 21]					
BLS2 (Chen & Liu, 2017)	[25, 0, 5]	[18, 2, 10]	[26, 0, 4]				
SVM2K (Farquhar et al., 2005)	[26, 0, 4]	[19, 1, 10]	[25, 0, 5]	[14, 0, 16]			
MVLDM (Hu et al., 2025)	[20, 3, 7]	[9, 0, 21]	[20, 1, 9]	[6, 0, 24]	[9, 0, 21]		
MvTSVM (Xie & Sun, 2015)	[11, 0, 19]	[5, 0, 25]	[12, 0, 18]	[4, 0, 26]	[2, 0, 28]	[6, 0, 24]	
CPBS-MvBLS ^a	[27, 0, 3]	[22, 1, 7]	[25, 1, 4]	[10, 2, 18]	[16, 1, 13]	[23, 0, 7]	[28, 0, 2]

^a Represents the proposed model.

Table 7

Wilcoxon signed-rank test on AwA datasets for the proposed CPBS-MvBLS model and baseline models.

CPBS-MvBLS (Proposed) ↓ Baseline →	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1(Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)
p-value	4.46E-06	0.011	1.36E-05	0.873	0.309	3.30E-04	6.34E-06
Null Hypothesis	Rejected	Rejected	Rejected	Not Rejected	Not Rejected	Rejected	Rejected

rior performance in terms of accuracy while maintaining a reasonable model complexity. Consequently, these experiments strengthen the empirical validation of the proposed CPBS-MvBLS model for practical applications.

6. Sensitivity analyses of the hyperparameters

This section examines the impact of hyperparameter variations on the performance of the proposed models, offering insights into how parameter selection influences robustness and adaptability across diverse datasets. By systematically tuning these hyperparameters, we identify ranges that maximize model performance and ensure stable, consistent results across different data scenarios.

6.1. Sensitivity analyses of hyperparameter μ on CPBS-BLS and CPBS-MvBLS models

By adjusting the parameter μ inside the range $\{2^{-5}, 2^{-4}, \dots, 2^5\}$ and fixing other parameters at their optimal values, we examined the effect of the kernel parameter μ on the performance of our proposed models. Fig. 3 illustrates the performance of the proposed CPBS-BLS model on the datasets: echocardiogram, haberman_survival, and the proposed CPBS-MvBLS model on the datasets: Persiencat vs Humpbackwhale, and Giantpanda vs Persiencat, achieving peak accuracy at different values of μ . In the echocardiogram dataset (Fig. 3(a)), the accuracy attains its optimal value on $\mu = 2^2$. The haberman_survival dataset (Fig. 3(b)) attains very low accuracy within the range $\{2^{-5}, 2^{-4}, \dots, 2^{-1}\}$, but increases drastically on $\mu = 1$ and achieves optimal accuracy on $\mu = 2^3$. Similarly, one can observe the Figs. 3(c) and (d). From the above analyses, we can observe that the performance of the proposed CPBS-BLS and CPBS-MvBLS models is affected by the different values of μ . Our result highlights the influence of dataset characteristics on the accuracy of the proposed CPBS-BLS and CPBS-MvBLS models, emphasizing the necessity of tuning the parameter μ to achieve optimal performance.

6.2. Sensitivity analyses of hyperparameters μ and τ on CPBS-BLS and CPBS-MvBLS models

In this subsection, we investigate the sensitivity of the hyperparameters by adjusting the values of μ and τ to analyze the effect of hyperparameters on the proposed CPBS-BLS and CPBS-MvBLS models' generalization capability. Fig. 4 illustrate the models' accuracies changes as the hyperparameters μ and τ are adjusted. It can be observed from Fig. 4(a) that the aus dataset achieves optimal accuracy when μ lies within the range $\{2^{-5}, 2^{-4}, \dots, 2^{-1}\}$ and τ is set to either $\frac{7}{8}$ or 1. As shown in Fig. 4(b), the echocardiogram dataset achieves optimal accuracy when μ is in the range $\{2^1, 2^2, \dots, 2^4\}$ and τ varies between $\frac{3}{4}$ and 1. Similarly, Figs. 4(c) and (d) (on CPBS-MvBLS) can be observed. Consequently, the observations emphasize the importance of carefully selecting hyperparameters μ and τ .

6.3. Sensitivity analyses of hyperparameter τ on CPBS-BLS and CPBS-MvBLS models

This subsection investigates the effect of the hyperparameter τ on the classification accuracy of the proposed models. As illustrated in Fig. 5,

models performance exhibit notable sensitivity to variations in τ . Specifically, Fig. 5(a) demonstrates that the highest accuracy is attained when $\tau = \frac{3}{4}$, whereas Fig. 5(b) indicates two optimal values at $\tau = \frac{5}{8}$ and $\tau = \frac{3}{4}$. For the CPBS-MvBLS model, Fig. 5(c) shows that the optimal accuracy is achieved at $\tau = \frac{7}{8}$, while Fig. 5(d) identifies $\tau = \frac{3}{4}$ and $\tau = 1$ as the most optimal values to achieve the optimal accuracy. Collectively, these findings highlight the critical role of appropriately selecting the hyperparameter τ to ensure optimal performance of the proposed models.

6.4. Sensitivity analyses of hyperparameters c (c_1) and n_Z on CPBS-BLS and CPBS-MvBLS models

Fig. 6 illustrate the sensitivity of the models accuracy to the number of feature groups (n_Z) and the regularization parameter c (or c_1). In Fig. 6(a), the highest accuracy is achieved when $n_Z \in \{30, \dots, 50\}$ and $c \in \{10^{-3}, \dots, 10^1\}$. A similar range of n_Z yields optimal results in Fig. 6(b), although the optimal values of c shift to the range $\{10^1, \dots, 10^5\}$. Figs. 6(c) and (d) also indicate peak accuracy within $n_Z \in \{35, \dots, 50\}$, with corresponding optimal values of c_1 in $\{10^1, \dots, 10^4\}$ and $\{10^{-3}, \dots, 10^0\}$, respectively. The variability observed in the optimal values of c (or c_1) underscores the greater sensitivity of the regularization parameter across different datasets.

6.5. Sensitivity analyses of hyperparameter n_Z on CPBS-BLS and CPBS-MvBLS models

To explore the impact of the hyperparameter n_Z on models performance, we conduct a comprehensive sensitivity analysis by evaluating different values of n_Z across multiple datasets. As illustrated in Fig. 7, the results demonstrate a clear relationship for the choice of n_Z to the optimal classification accuracy. Figs. 7(a) and (b) show that the model's accuracy improves monotonically with increasing values of n_Z , reaching its peak performance at $n_Z = 45$ and $n_Z = 30$, respectively. Similarly, Figs. 7(c) and (d) exhibit analogous behavior, with the highest accuracy achieved at $n_Z = 50$ and $n_Z = 40$, respectively. These findings underscore the importance of carefully tuning n_Z to strike a balance between model generalization capabilities.

7. Ablation study on weighting schemes

To demonstrate the contribution of the proposed CPBS weighting function, we conducted an ablation study in which the weighting membership function in CPBS-BLS and CPBS-MvBLS was systematically replaced by alternative membership functions while keeping all other hyperparameters and data splits fixed. Specifically, we compared: a linear weighting, an exponential weighting, no weighting, and the proposed CPBS weighting. The classification accuracy (%) is used as an evaluation metric in all benchmark data sets. Tables 12 and 13 summarize the results. For the proposed CPBS-BLS, the CPBS weighting achieves an average accuracy of 89.47%, which is substantially higher than the linear (74.03%), exponential (69.44%), and no weighting (67.15%). The inferior performance of linear and exponential weightings stems from their reliance on a fixed single criterion (distance), without considering local class density or class probability. As a result, noisy, borderline, or sparsely located samples may still receive relatively high weight. In contrast, CPBS adaptively integrates distance, density, and imbalance

Table 8

Performance comparison of baseline and proposed CPBS-MvBLS model on UCI and KEEL datasets.

Dataset ↓ Model →	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)	CPBS-MvBLS ^a
Best hyperparameters	Acc. ± Std. (c_1, n_H)	Acc. ± Std. (c_1, n_H)	Acc. ± Std. (c_1, n_Z, m_Z, n_H)	Acc. ± Std. (c_1, n_Z, m_Z, n_H)	Acc. ± Std. (c_1, μ)	Acc. ± Std. (c_1, c_2, c_3, μ)	Acc. ± Std. (c_1, c_2, μ)	Acc. ± Std. (c_1, n_Z, m_Z, n_H, τ)
abalone9-18	95 ± 5.88 (1, 163)	96.82 ± 4.82 (10 ² , 83)	95.91 ± 9.61 (100, 10, 5, 10)	96.36 ± 12.53 (10 ⁻⁴ , 20, 1, 20)	94.52 ± 1.54 (16, 0.125)	93.15 ± 1.56 (2, 0.25, 0.25, 4)	95.7 ± 16.75 (10 ⁻² , 10 ⁻¹ , 10 ⁻⁴)	95.45 ± 0.54 (10 ⁻⁴ , 45, 3, 45, 2)
acute_inflammation	83.33 ± 0 (10 ⁻⁴ , 63)	97.22 ± 0 (10 ⁻⁴ , 203)	100 ± 0 (10 ⁻⁴ , 5, 5, 5)	100 ± 0 (10 ⁻⁴ , 10, 5, 10)	100 ± 0 (0.125, 1)	100 ± 13.97 (0.25, 0.25, 0.25, 0.25)	90.48 ± 0 (10 ⁻² , 10 ⁻³ , 1)	97.22 ± 0 (10 ⁻⁴ , 5, 3, 5, 0.125)
acute_nephritis	100 ± 0 (10 ⁻³ , 103)	100 ± 0 (10 ⁻² , 63)	91.67 ± 0 (10 ⁻⁴ , 5, 19, 5)	100 ± 0 (10 ⁻⁴ , 5, 21, 5)	100 ± 0 (0.125, 0.5)	100 ± 0 (0.25, 0.25, 0.25, 0.25)	100 ± 0 (10 ⁻⁴ , 0.1, 1)	100 ± 0 (10 ⁻⁴ , 5, 3, 5, 0.25)
aus	89.9 ± 4.9 (10 ⁻² , 163)	88.46 ± 4.9 (10 ⁻¹ , 123)	87.98 ± 5.88 (10 ⁻¹ , 15, 17, 15)	86.06 ± 5.68 (10 ⁻¹ , 50, 7, 50)	81.16 ± 2.8 (0.125, 32)	86.53 ± 3.85 (0.25, 0.25, 4, 2)	83.44 ± 6.63 (10 ⁻³ , 10 ⁻³ , 10 ⁻³)	87.92 ± 1.71 (1, 30, 9, 30, 4)
blood	76.89 ± 8.41 (100, 23)	76 ± 8.08 (0.1, 183)	77.33 ± 2.23 (1, 20, 17, 20)	76.11 ± 3.32 (0.1, 50, 9, 50)	78.57 ± 2.44 (1, 0.5)	80.8 ± 8.09 (1, 1, 0.25, 0.25)	75.57 ± 2.98 (10 ⁻² , 10 ⁻¹ , 10 ⁻⁴)	77.33 ± 18.41 (0.1, 5, 13, 5, 0.03125)
breast_cancer_wisc	95.71 ± 2.53 (10 ⁻¹ , 63)	96.19 ± 2.53 (10 ⁻¹ , 163)	88.1 ± 24.95 (10, 10, 1, 10)	93.1 ± 34.11 (10, 20, 5, 20)	91.61 ± 5.91 (0.0625, 0.0625)	93.61 ± 8.23 (4, 4, 0.25, 0.25)	89.9 ± 13.55 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	94.76 ± 2.49 (10 ⁻³ , 40, 17, 40, 0.5)
breast_cancer_wisc_diag	93.57 ± 3.79 (10 ⁻² , 163)	91.81 ± 3.79 (10 ⁻¹ , 203)	89.42 ± 3.97 (1, 15, 13, 15)	87.08 ± 7.74 (10, 5, 9, 5)	95.49 ± 6.98 (8, 8)	93.15 ± 9.37 (0.5, 0.5, 0.25, 2)	92.98 ± 5.95 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	97.66 ± 4.6 (10 ⁻³ , 25, 15, 25, 2)
breast_cancer_wisc_prog	78.33 ± 1.82 (10 ⁻¹ , 83)	23.33 ± 1.82 (10 ⁻³ , 203)	68.33 ± 11.36 (10, 25, 19, 25)	68.33 ± 12.69 (10 ² , 15, 13, 15)	58.33 ± 9.21 (2, 2)	71.17 ± 15.4 (0.25, 0.25, 0.25, 0.25)	74.82 ± 9.69 (10 ⁻² , 10 ⁻¹ , 10 ⁻⁴)	81.96 ± 12.4 (10 ⁻³ , 5, 1, 5, 8)
brwiscosin	94.63 ± 1.25 (10 ⁻² , 63)	95.12 ± 5.88 (10 ⁻³ , 143)	91.07 ± 14.92 (1, 40, 3, 40)	96.1 ± 3.03 (10, 40, 15, 40)	98.04 ± 6.56 (0.5, 0.5)	95.59 ± 1.71 (0.25, 0.25, 0.25, 0.25)	95.82 ± 4.78 (10 ⁻³ , 10 ⁻³ , 10 ⁻³)	96.45 ± 1.51 (10 ⁻⁴ , 5, 3, 5, 16)
bupa_or_liver_disorders	52.88 ± 6.93 (0.1, 163)	49.04 ± 6.93 (10 ² , 23)	63.46 ± 5.88 (1, 50, 5, 50)	65.38 ± 12.51 (1, 20, 17, 20)	65.05 ± 5.4 (0.5, 0.5)	55.34 ± 6.81 (0.5, 0.5, 0.25, 0.25)	55.37 ± 7.81 (10 ⁻³ , 10 ⁻³ , 10 ⁻³)	72.63 ± 7.88 (10 ³ , 5, 7, 5, 2)
checker-board_Data	89.9 ± 1.82 (10 ⁻² , 163)	88.46 ± 1.82 (10 ⁻¹ , 123)	83.98 ± 8.6 (10 ⁻¹ , 15, 17, 15)	86.06 ± 5.68 (10 ⁻¹ , 50, 7, 50)	81.16 ± 4.47 (8, 8)	84.06 ± 13.3 (0.25, 0.25, 0.25, 2)	81.16 ± 7.81 (10 ⁻³ , 10 ⁻³ , 10 ⁻³)	86.54 ± 3.84 (10 ⁻¹ , 40, 7, 40, 16)
credit_approval	84.13 ± 6.99 (10 ⁻² , 83)	84.13 ± 6.99 (10 ² , 63)	82.62 ± 3.46 (10 ⁻¹ , 20, 11, 20)	84.62 ± 3.34 (1, 5, 15, 5)	85.99 ± 5 (0.125, 0.125)	85.51 ± 4.62 (4, 4, 0.25, 0.25)	79.3 ± 3.64 (10 ⁻¹ , 10 ⁻³ , 1)	83.65 ± 3.51 (10 ⁻² , 45, 7, 45, 2)
cylinder_bands	70.13 ± 4.9 (100, 143)	66.23 ± 4.9 (10 ⁻² , 203)	71.62 ± 3.44 (1, 45, 5, 45)	74.03 ± 3.66 (10 ⁻² , 35, 7, 35)	73.2 ± 4.42 (16, 8)	71.9 ± 5.65 (2, 2, 0.25, 0.25)	73.26 ± 18.48 (10 ⁻³ , 10 ⁻³ , 10 ⁻³)	72.61 ± 2.56 (10 ² , 5, 11, 5, 2)
echocardiogram	80 ± 6.78 (10 ⁻¹ , 103)	85 ± 6.78 (10 ⁻² , 203)	82.5 ± 7.46 (10, 40, 19, 40)	87.5 ± 5.81 (10 ⁻⁴ , 35, 21, 35)	79.49 ± 8.05 (0.5, 8)	84.62 ± 5.99 (0.25, 0.25, 0.25, 0.25)	76.09 ± 3.19 (10 ⁻² , 10 ⁻³ , 1)	87.5 ± 6.51 (100, 5, 17, 5, 32)
fertility	86.67 ± 5.45 (10 ⁻¹ , 63)	80 ± 5.45 (10 ⁻¹ , 83)	90 ± 12.5 (10 ⁴ , 5, 1, 5)	100 ± 9.58 (100, 5, 1, 5)	86.67 ± 10.83 (0.03125, 0.03125)	86.67 ± 10.83 (0.25, 0.25, 0.25, 0.25)	88.57 ± 6.78 (10 ⁻³ , 10 ⁻³ , 10 ⁻³)	90 ± 11.73 (10 ⁻⁴ , 30, 11, 30, 8)
haberman_survival	78.26 ± 8.08 (10 ⁻² , 143)	79.35 ± 2.63 (10 ⁻² , 123)	76.09 ± 7.37 (10 ⁻³ , 10, 9, 10)	78.26 ± 8.47 (10 ⁻³ , 45, 1, 45)	74.73 ± 9.94 (4, 2)	71.43 ± 2.84 (0.25, 0.25, 0.25, 0.25)	71.16 ± 12.18 (10 ⁻⁴ , 10 ⁻³ , 10 ⁻¹)	80.43 ± 5.64 (10 ⁻² , 10, 1, 10, 0.125)
horse_colic	76.58 ± 4.02 (10 ⁻² , 163)	76.58 ± 4.02 (10 ⁻² , 163)	80.78 ± 14.28 (10, 15, 11, 15)	84.68 ± 6.51 (0.1, 50, 1, 50)	81.82 ± 11.9 (2, 32)	83.64 ± 4.18 (0.25, 0.25, 0.25, 4)	77.91 ± 7.71 (10 ⁻³ , 10 ⁻⁴ , 1)	81.3 ± 6.38 (10 ⁻⁴ , 20, 9, 20, 0.03125)
ilpd_indian_liver	70.29 ± 8.92 (1, 83)	58.86 ± 8.92 (10 ³ , 3)	68.57 ± 2.3 (10 ² , 30, 7, 30)	70.29 ± 11.67 (10 ² , 25, 9, 25)	72.99 ± 8.36 (1, 0.03125)	72.99 ± 6.58 (2, 2, 0.25, 0.25)	71.15 ± 7.11 (100, 0.1, 10 ⁻³)	71.09 ± 3.46 (10 ⁻⁴ , 5, 11, 5, 32)
mammographic	78.89 ± 9.58 (10 ⁴ , 63)	76.47 ± 9.58 (1, 83)	70.28 ± 2.48 (10 ⁻⁴ , 20, 7, 20)	72.01 ± 4.02 (10 ⁻⁴ , 25, 1, 25)	82.29 ± 7.98 (16, 4)	80.9 ± 4.12 (4, 4, 0.25, 2)	81.58 ± 9.77 (10 ⁻³ , 10 ⁻³ , 10 ⁻³)	79.93 ± 4.8 (10 ⁻¹ , 20, 1, 20, 8)
molec_biol_promoter	71.88 ± 3.84 (10 ³ , 163)	59.38 ± 3.84 (10 ³ , 203)	56.25 ± 8.28 (10 ⁻² , 50, 17, 50)	81.25 ± 11.8 (10 ⁻¹ , 30, 21, 30)	83.87 ± 6.1 (1, 32)	83.87 ± 1.85 (0.25, 0.25, 0.25, 0.25)	73.33 ± 11.92 (0.1, 10 ⁻² , 10 ⁻⁵)	81.25 ± 9.9 (10 ⁻¹ , 25, 19, 25, 4)

Table 8
Continue.

Dataset ↓ Model →	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)	CPBS-MvBLS ^a
shuttle-6_vs_2-3	100 ± 6.05 (10 ⁻⁴ , 23)	100 ± 6.057 (23, 2)	100 ± 3.09 (10, 40, 15, 40)	100 ± 12.42 (10, 15, 7, 15)	97.1 ± 19.01 (0.03125, 0.03125)	95.65 ± 2.21 (0.25, 0.25, 0.25, 0.25)	96 ± 3.12 (10 ⁻⁴ , 0.1, 1)	100 ± 6.2 (10 ⁻⁴ , 50, 17, 50, 1)
sonar	65.08 ± 7.31 (1, 123)	63.49 ± 7.31 (203, 2)	77.78 ± 11.97 (10 ⁻¹ , 40, 5, 40)	79.37 ± 4.19 (10 ⁻² , 20, 13, 20)	85.48 ± 8.12 (2, 0.25)	79.03 ± 7.42 (4, 4, 0.25, 4)	79.45 ± 4.3 (10 ⁻² , 1, 0.1)	81.38 ± 7.47 (10 ⁻³ , 5, 13, 5, 32)
statlog_aus-tralian_credit	53.85 ± 7.82 (10 ⁻⁴ , 3)	68.75 ± 7.82 (3, 2)	70.67 ± 10.33 (10 ³ , 5, 17, 5)	70.67 ± 12.33 (10 ⁻³ , 5, 21, 5)	69.57 ± 6.66 (0.5, 0.5)	71.98 ± 3.97 (0.25, 0.25, 4, 0.25)	67.08 ± 11.3 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	69.71 ± 5.12 (10 ⁻⁴ , 5, 1, 5, 4)
statlog_ger-man_credit	77.33 ± 6.24 (10 ⁻² , 83)	76 ± 6.24 (203, 2)	76.33 ± 3.08 (10 ⁻¹ , 10, 9, 10)	78 ± 6.22 (1, 35, 7, 35)	74 ± 6.98 (4, 16)	73.67 ± 1.79 (1, 1, 0.25, 0.25)	69.57 ± 2.54 (10 ⁻¹ , 10 ⁻³ , 1)	77.33 ± 4.47 (100, 20, 19, 20, 8)
statlog_heart	83.95 ± 3.09 (10 ⁻⁴ , 63)	90.12 ± 8.41 (163, 2)	82.72 ± 2.11 (10 ⁻² , 5, 5, 5)	81.89 ± 3.56 (10 ⁻³ , 45, 7, 45)	82.72 ± 5.47 (0.5, 16)	82.72 ± 2.36 (0.25, 0.25, 0.25, 0.25)	73.54 ± 6.36 (100, 0.1, 10 ⁻³)	87.65 ± 8.8 (0.1, 50, 1, 50, 1)
tic_tac_toe	99.65 ± 5.73 (10, 143)	81.6 ± 5.73 (83, 2)	89.65 ± 1.28 (0.1, 50, 15, 50)	70.49 ± 7.91 (1, 50, 1, 50)	90 ± 7.27 (4, 4)	90 ± 3.83 (0.25, 0.25, 0.25, 0.25)	98.06 ± 1.39 (10 ⁻³ , 10 ⁻⁴ , 1)	96.88 ± 7.5 (10 ⁻⁴ , 10, 5, 10, 2)
transfusion	76.89 ± 8.58 (10 ⁻² , 103)	76 ± 8.58 (3, 2)	75.11 ± 11.84 (1, 20, 17, 20)	76.89 ± 9.47 (10, 5, 13, 5)	76.34 ± 5.62 (0.03125, 0.03125)	75 ± 2.77 (0.5, 0.5, 0.25, 0.5)	77.1 ± 4.6 (10 ⁻² , 10 ⁻⁴ , 10 ⁻¹)	81.05 ± 3.33 (1, 10, 19, 10, 0.03125)
vehicle1	78.74 ± 7.13 (100, 163)	72.05 ± 7.13 (23, 2)	79.13 ± 8.1 (0.1, 10, 3, 10)	82.28 ± 10.07 (1, 40, 15, 40)	75.89 ± 4.62 (1, 32)	65.61 ± 2.87 (0.25, 0.25, 0.25, 0.25)	81.11 ± 4.84 (10 ⁻² , 10 ⁻¹ , 10 ⁻⁴)	79.22 ± 0.75 (10 ⁻³ , 5, 1, 5, 8)
votes	92.37 ± 4.64 (0.1, 183)	96.95 ± 4.61 (23, 2)	92.42 ± 16.35 (100, 20, 7, 20)	90.47 ± 2.12 (10, 25, 19, 25)	93.08 ± 4.7 (4, 32)	90.58 ± 3.01 (0.25, 0.25, 0.25, 0.25)	93.77 ± 13.38 (10 ⁻⁴ , 10 ⁻⁴ , 10 ⁻⁴)	97.71 ± 0 (0.1, 10, 3, 10, 1)
wdbc	35.59 ± 8.81 (10 ² , 23)	35.59 ± 8.81 (143, 2)	77.97 ± 5.28 (100, 40, 19, 40)	79.66 ± 7.68 (100, 25, 21, 25)	72.41 ± 11.8 (2, 1)	61.01 ± 3.81 (2, 2, 0.25, 0.25)	75.74 ± 8.96 (10 ⁻² , 10 ⁻³ , 1)	82.22 ± 3.4 (10 ⁻³ , 5, 13, 5, 32)
Average accuracy	80.35 ± 5.37	77.63 ± 5.48	81.26 ± 7.41	83.23 ± 7.94	82.72 ± 6.59	82.01 ± 5.43	81.3 ± 7.26	85.63 ± 5.16

^a Represents the proposed model. Boldface highlights the model with the highest average accuracy. Avg. and Std. are acronyms of average and standard deviation.

Table 9

Win-tie-loss test on UCI and KEEL datasets for the proposed CPBS-MvBLS model.

	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)
ELM2 (Huang et al., 2006)	[9, 5, 16]						
BLS1 (Chen & Liu, 2017)	[13, 1, 16]	[14, 1, 15]					
BLS2 (Chen & Liu, 2017)	[17, 2, 11]	[18, 2, 10]	[20, 4, 6]				
SVM2K (Farquhar et al., 2005)	[15, 1, 14]	[19, 1, 10]	[16, 1, 13]	[11, 2, 17]			
MVLDM (Hu et al., 2025)	[14, 1, 15]	[17, 1, 12]	[17, 1, 12]	[13, 2, 15]	[8, 7, 15]		
MvTSVM (Xie & Sun, 2015)	[16, 1, 13]	[15, 1, 14]	[14, 0, 16]	[8, 1, 21]	[10, 2, 18]	[14, 1, 15]	
CPBS-MvBLS ^a	[22, 2, 6]	[22, 2, 6]	[23, 3, 4]	[17, 4, 9]	[19, 1, 10]	[21, 1, 8]	[23, 1, 6]

^a Represents the proposed model.

Table 10

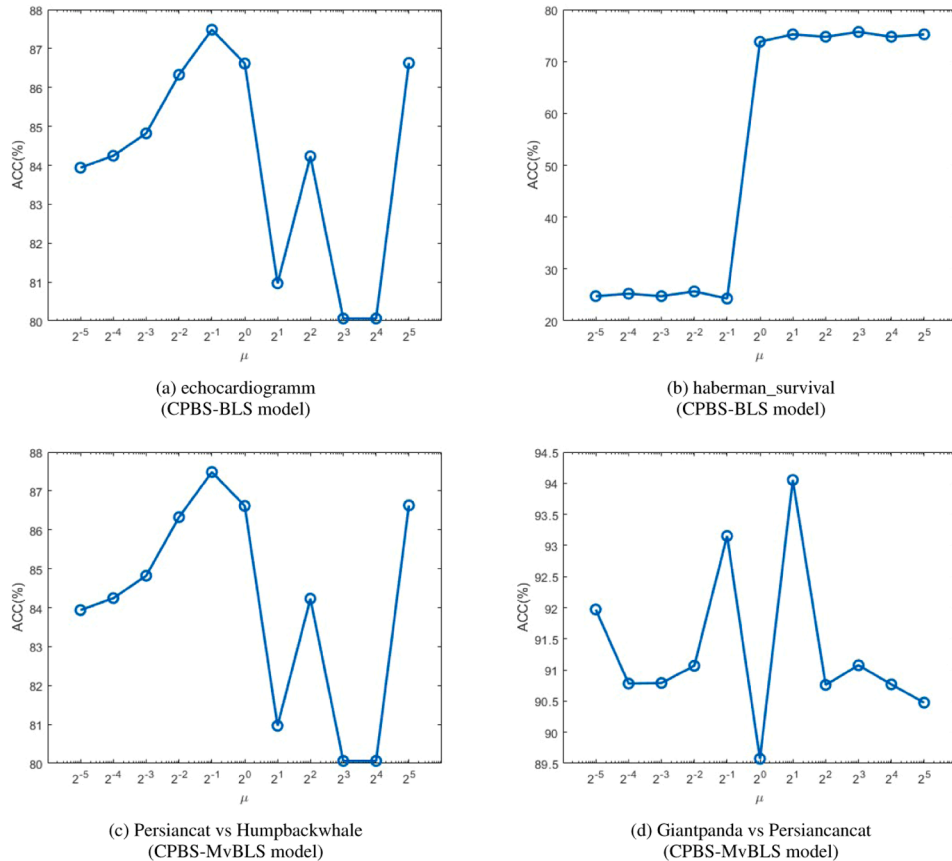
Wilcoxon signed-rank test on UCI and KEEL dataset for the proposed CPBS-MvBLS model and baseline models.

CPBS-MvBLS (Proposed) ↓ Baseline →	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)
p-value	5.17E-04	5.40E-04	4.90E-05	0.058	0.003	0.002	4.17E-05
Null Hypothesis	Rejected	Rejected	Rejected	Not Rejected	Rejected	Rejected	Rejected

Table 11

Performance comparison of baseline and proposed CPBS-MvBLS model on MNIST datasets.

Dataset name ↓ Model →	ELM1 (Huang et al., 2006)	ELM2 (Huang et al., 2006)	BLS1 (Chen & Liu, 2017)	BLS2 (Chen & Liu, 2017)	SVM2K (Farquhar et al., 2005)	MVLDM (Hu et al., 2025)	MvTSVM (Xie & Sun, 2015)	CPBS-MvBLS ^a
Best Hyper- parameter →	(c_1, n_H)	(c_1, n_H)	(c_1, n_Z, m_Z, n_H)	(c_1, n_Z, m_Z, n_H)	(c_1, μ)	(c_1, c_2, c_3, μ)	(c_1, c_2, μ)	$(c_1, n_Z, m_Z, n_H, \tau)$
0vs1	96.56 (100, 83)	89.9 (10^2 , 203)	81.1 (10, 20, 3, 20)	93.76 (10^3 , 20, 17, 20)	95.88 (0.5, 32)	94.76 (0.25, 0.25, 0.25, 0.25)	88.4 (0.03125, 0.03125, 16)	98.27 (1, 20, 1, 20, 1)
1vs2	89.18 (100, 163)	91.33 (10^{-2} , 203)	80.15 (1, 40, 17, 40)	58.48 (10^4 , 10, 17, 10)	99.7 (2, 32)	93.94 (0.25, 0.25, 0.25, 4)	95.85 (0.03125, 8)	97.85 (10^{-2} , 50, 5, 50, 0.25)
2vs3	79.28 (10^2 , 183)	98.38 (10^{-1} , 183)	97.52 (10^{-1} , 10, 17, 10)	96.95 (0.1, 25, 21, 25)	87.24 (4, 32)	96.24 (1, 0.25, 0.25, 4)	92.43 (0.03125, 32)	97.55 (0.1, 25, 17, 25, 4)
2vs5	98.94 (100, 183)	98.1 (10^{-1} , 163)	97.97 (0.1, 40, 11, 40)	91.29 (0.1, 45, 9, 45)	84.19 (0.5, 16)	98.99 (0.25, 0.25, 0.25, 4)	90.59 (0.03125, 32)	97.56 (10^{-2} , 15, 21, 15, 0.125)
2vs7	88.5 (100, 203)	98.3 (10^{-1} , 103)	90.52 (1, 10, 11, 10)	98.06 (0.1, 10, 19, 10)	99.18 (4, 32)	89.03 (0.25, 0.25, 4, 4)	98.45 (0.03125, 32)	99.25 (0.1, 15, 5, 15, 0.5)
3vs4	98.78 (10, 183)	76.54 (10, 183)	99.8 (0.1, 5, 21, 5)	99.6 (0.1, 40, 19, 40)	99.6 (2, 16)	99.6 (0.25, 0.25, 0.25, 0.25)	99.8 (0.03125, 32)	98.8 (0.1, 5, 21, 5, 0.125)
5vs6	94.1 (10^{-2} , 203)	93.82 (10, 183)	92.04 (1, 30, 3, 30)	84.3 (0.1, 15, 13, 15)	91.78 (2, 16)	99.35 (0.25, 0.25, 0.25, 4)	97.85 (0.03125, 32)	96.31 (0.1, 5, 21, 5, 4)
Average Accuracy	92.19	92.33	91.3	88.92	93.93	95.98	94.76	97.94

^a Represents the proposed model.**Fig. 3.** Effect of the hyperparameter μ on the accuracy of the proposed CPBS-BLS and CPBS-MvBLS models.

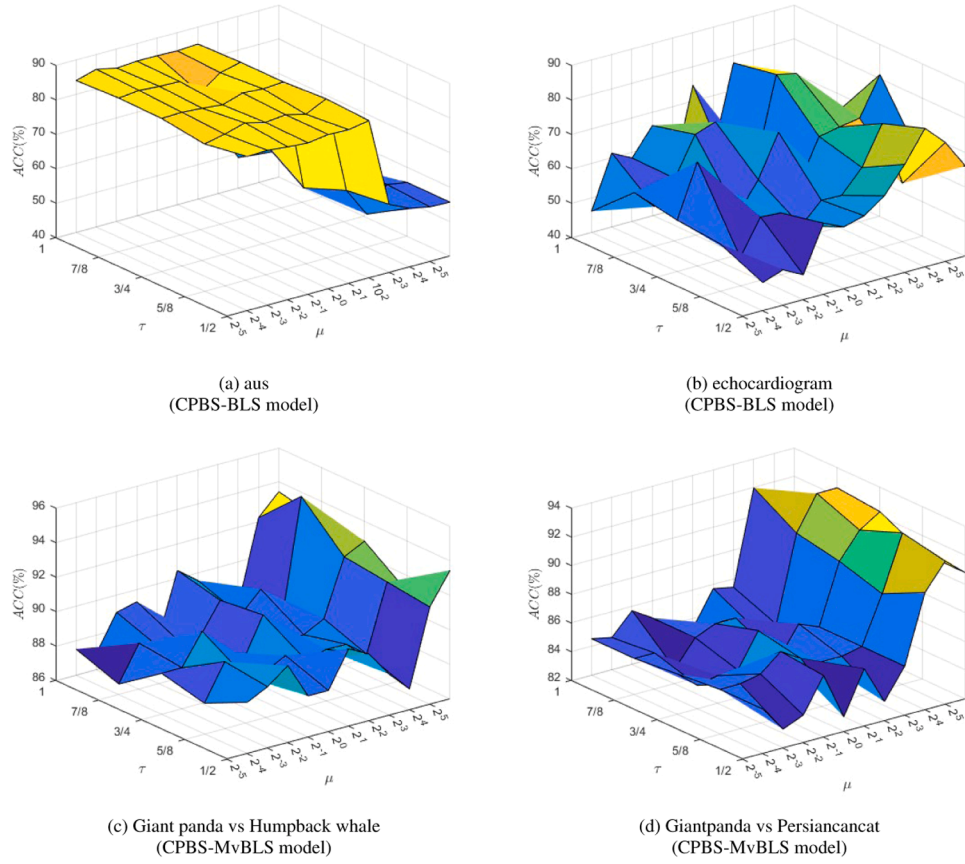


Fig. 4. Effect of the hyperparameters μ and τ on the accuracy of the proposed CPBS-BLS and CPBS-MvBLS models.

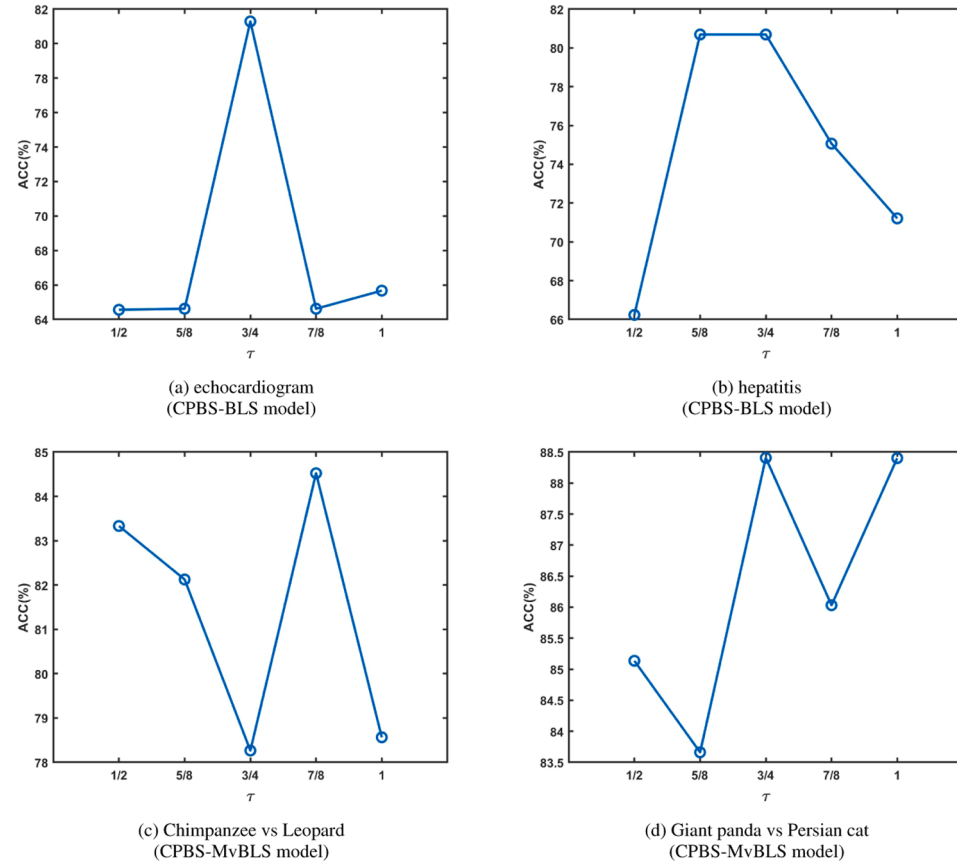


Fig. 5. Effect of the hyperparameter τ on the accuracy of the proposed CPBS-BLS and CPBS-MvBLS models.

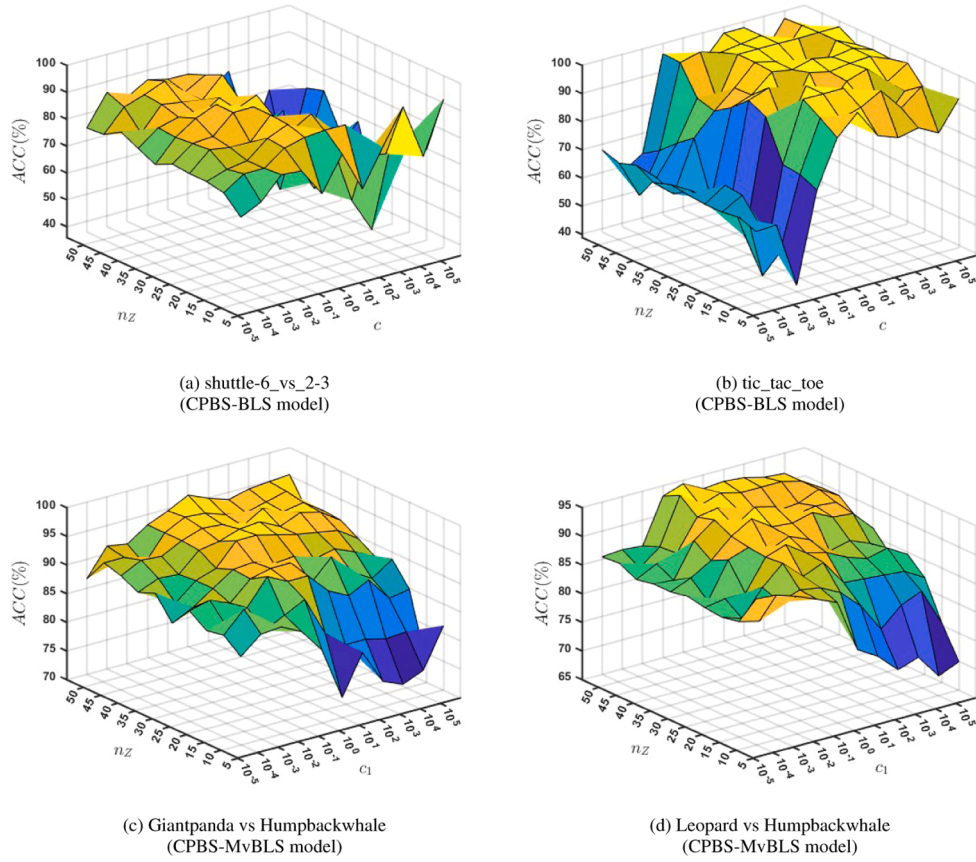


Fig. 6. Effect of the hyperparameters c or c_1 and n_Z on the accuracy of the proposed CPBS-BLS and CPBS-MvBLS models.

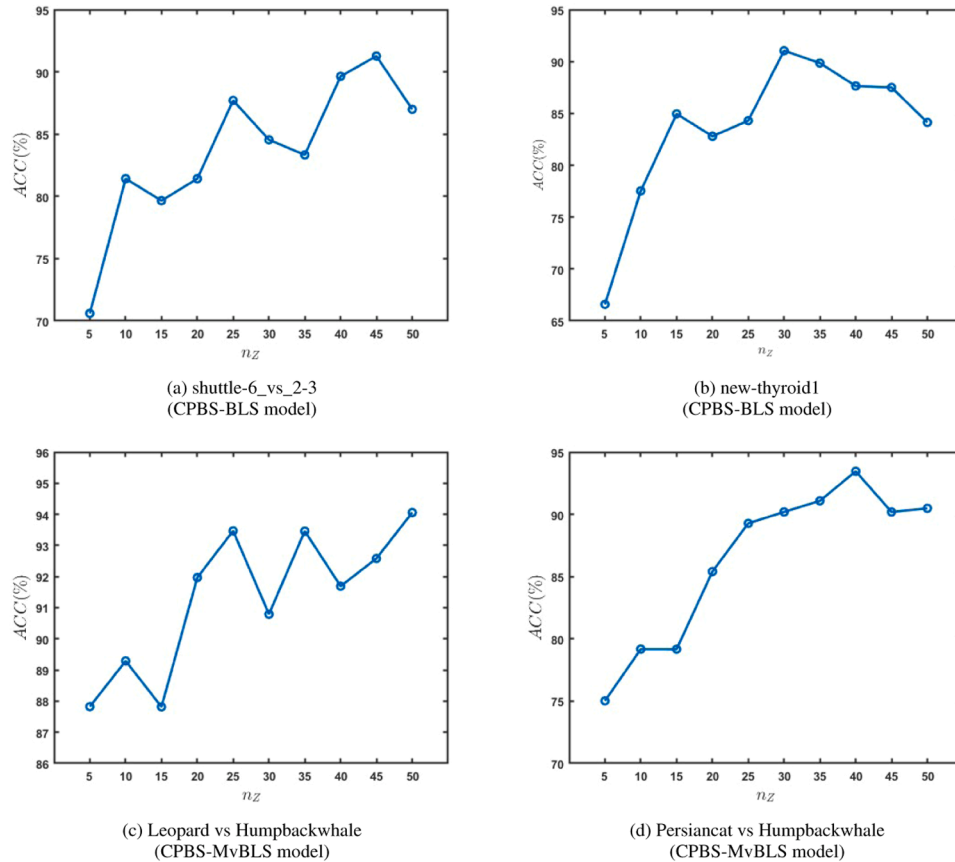


Fig. 7. Effect of the hyperparameters n_Z on the accuracy of the proposed CPBS-BLS and CPBS-MvBLS models.

Table 12

Ablation study on the proposed CPBS-BLS model on UCI and KEEL datasets.

Dataset ↓ Model →	CPBS-BLS (Linear)	CPBS-BLS (Exponential)	BLS (No weighting)	CPBS-BLS ^a
echocardiogram	64.93	62.54	55.54	87.5
haberman_survival	77.43	66.32	77.01	79.35
hepatitis	89.25	80.15	76.19	80.85
shuttle-6_vs_2-3	71.87	72.5	63.75	100
tic_tac_toe	66.67	65.67	63.28	99.65
Average Accuracy	74.03	69.44	67.15	89.47

^a Represents the proposed model.**Table 13**

Ablation study on the proposed CPBS-MvBLS model on AWA, UCI, and KEEL datasets.

Dataset ↓ Model →	CPBS-MvBLS (Linear)	CPBS-MvBLS (Exponential)	CPBS-MvBLS ^a
Persiancat vs Humpbackwhale	79.17	79.17	91.67
Giantpanda vs Persiancat	65.28	65.28	84.72
Leopard vs Humpbackwhale	79.17	77.08	90.97
echocardiogram	75	77.5	87.5
haberman_survival	78.26	78.26	80.43
hepatitis	31.91	57.45	78.72
shuttle-6_vs_2-3	86.96	86.96	100
tic_tac_toe	73.62	81.60	96.88
Average Accuracy	71.17	75.41	88.86

^a Represents the proposed model.**Table 14**Ablation study evaluating the impact of the coupling parameter (ρ) on CPBS-MvBLS performance across AWA, UCI, and KEEL datasets.

Dataset ↓ Model →	CPBS-MvBLS (without ρ)	CPBS-MvBLS ^a
Persiancat vs Humpbackwhale	90.36	91.67
Giantpanda vs Persiancat	77.78	84.72
Leopard vs Humpbackwhale	91.36	90.97
echocardiogram	90	87.5
haberman_survival	71.74	80.43
hepatitis	76.6	78.72
shuttle-6_vs_2-3	100	100
tic_tac_toe	93.65	96.88
Average Accuracy	85.41	88.86

^a Represents the proposed model.

information, leading to more reliable sample contribution and superior accuracy. For the proposed CPBS-MvBLS, the proposed weighting achieves an average accuracy of 88.86%, outperforming both the linear (71.17%) and exponential (75.41%) weightings. These results clearly indicate that the CPBS weighting function itself plays a crucial role. By assigning larger weights to informative samples while down-weighting potentially noisy or outlying samples.

Furthermore, Table 14 presents an ablation study analyzing the effect of the coupling parameter ρ on the performance of the CPBS-MvBLS model across datasets from AWA, UCI, and KEEL. The results demonstrate that incorporating the coupling parameter consistently improves overall performance. Specifically, the average accuracy increases from 85.41% to 88.86%, confirming the effectiveness of the coupling parameter in enhancing the proposed CPBS-MvBLS model generalization. A dataset-wise comparison further highlights the benefits of incorporating ρ . For example, the accuracy on Giantpanda vs Persiancat increases from 77.78% to 84.72%, while the haberman-survival dataset improves substantially from 71.74% to 80.43%. Similarly, improvements are observed on Persiancat vs Humpbackwhale, hepatitis, and tic_tac_toe, demonstrating the effectiveness of the coupling mechanism in leveraging complementary information from different views. The overall results indicate that introducing the coupling parameter ρ in the Eq. 18 strengthens the learning capability of CPBS-MvBLS by better capturing inter-view relationships, thereby improving classification accuracy across most datasets. This coupling mechanism enables the model to better capture cross-view correlations and complementary patterns that may not be fully exploited when the views are treated

independently. Consequently, the CPBS-MvBLS framework learns more informative and discriminative feature representations, leading to improved classification accuracy and greater robustness across heterogeneous datasets.

8. Conclusion

This paper introduces two advanced broad learning system (BLS) variants: CPBS-BLS and CPBS-MvBLS, designed to address challenges posed by noisy, imbalanced, and complex real-world datasets. The CPBS-BLS model improves robustness to outliers and class imbalance by employing a bell-shaped membership function conditioned on class probability and imbalance ratio. This mechanism adaptively weights each sample according to its reliability, reducing the influence of noisy or atypical data points and enhancing generalization performance. Recognizing the limitations of single-view data in capturing intricate patterns, we further propose CPBS-MvBLS, a multi-view extension of CPBS-BLS. By integrating features from multiple perspectives, CPBS-MvBLS captures complementary information across different views, leading to richer data representations and improved predictive performance. The model also leverages the CPBS membership function to prioritize informative features while mitigating noise and inconsistencies across views. Extensive experiments on UCI, KEEL, and AWA datasets demonstrate that both CPBS-BLS and CPBS-MvBLS outperform traditional BLS frameworks in terms of accuracy, robustness, and stability. Comprehensive statistical evaluations—including ranking schemes, Friedman tests, Nemenyi post hoc tests, Win-tie-loss analyses, and wilcoxon signed rank test—confirm the superiority of the proposed models. Sensitivity analyses further emphasize the importance of careful hyperparameter tuning to maximize model performance. On average, CPBS-BLS achieves an improvement of approximately 3–5 percentage points in classification accuracy over strong baseline BLS variants across UCI and KEEL datasets, while CPBS-MvBLS further improves performance by about 2–3 percentage points over existing multi-view BLS models, demonstrating consistent and statistically significant gains across diverse and challenging benchmark datasets. The proposed models lay a foundation for future extensions to unsupervised and semi-supervised learning, where the scarcity of labeled data poses additional challenges. Developing an incomplete-view CPBS-MvBLS with view-completion and CPBS-based down-weighting of uncertain samples would improve applicability to heterogeneous data. In addition, the membership design can be improved through adaptive neighborhood

and kernel learning, particularly in high-dimensional and imbalanced settings.

CRedit authorship contribution statement

M. Tanveer: Writing – original draft, Supervision, Project administration, Conceptualization; **M. Pathak:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization; **M. Sajid:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization; **A. Quadir:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization; **Priyamvada:** Writing – original draft, Visualization, Validation, Supervision, Methodology.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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